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The Astrophysical Gravitational Wave Background Created by Coalescing Black Holes

The Effects of Eccentricity, Mass Transfer and Metallicity

Thesis BSc Physics

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Abstract

This work extends the theoretical framework developed by Staelens and Nelemans [2024] and further elaborated by Hofman and Nelemans [2024] for white dwarf binary systems to the case of coalescing black hole binaries, with the aim of characterising the astrophysical gravitational wave background (AGWB) produced by this population across a broad frequency range. The AGWB is the combined signal formed by the overlapping gravitational waves from all astrophysical sources throughout the universe, too numerous and too faint individually to be resolved as separate sources. The background is computed by summing the gravitational wave signal of each binary system per frequency bin and per redshift, integrating over the full cosmological population using a metallicity dependent star formation rate density. The framework accounts for two frequency regimes, the static and inspiral regimes, and three integration cases determined by the binary's position relative to its birth and merger frequencies. The circular orbit approximation is validated against an eccentricity correction computed for a test population, confirming that the correction factor is negligible across the frequency range of interest. How a binary evolves before it becomes a black hole pair, in particular how the two stars exchange mass and shed their outer layers, determines how close together the resulting black holes end up, which in turn sets how many of them merge within the age of the universe and thus how loud the background is. Two formalisms describing this mass and energy exchange, referred to as γ and α, are compared, together with five metallicity dependent star formation rate models. The results show that the choice between the γ and α formalism has a negligible impact on the background amplitude, while the efficiency with which orbital energy is used to eject the stars' outer envelopes has a modest but measurable effect, with higher efficiency producing a slightly lower amplitude by leaving more systems too widely separated to merge in time. Metallicity is identified as the dominant population parameter, as higher metallicity suppresses the formation of massive black hole binaries through stronger stellar winds, directly reducing the background via the $M_c^{5/3}$ chirp mass dependence. The resulting background sits on the high end of the LIGO Scientific Collaboration et al. [2025] estimated range, a factor of two to three above comparable studies using different population synthesis codes, which is attributed to the limited parameter space explored and to properties of SeBa that seems to give a higher survivavbility for higher mass binaries. Within the LISA sensitivity band, the black hole binary background follows the expected $f^{2/3}$ inspiral scaling and in most models falls below the white dwarf background, further supporting the conclusion that white dwarf binaries are the dominant source of gravitational waves in the LISA frequency range.

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1 Introduction

Throughout the universe, many interesting objects and processes can be found which may reveal the foundations of the universe, while raising questions we otherwise would not have asked. All matter in the universe has essentially been altered or created by burning engines called stars and the processes that come with them, making it a very interesting natural laboratory to study.

Stars form from the gravitational collapse of molecular gas clouds, mainly composed of hydrogen and helium. As a cloud collapses, it heats up and eventually ignites nuclear fusion in its core, entering what is known as the main sequence, the prolonged phase during which a star fuses hydrogen into helium and during which most stars spend the majority of their lives. The duration of the main sequence is determined primarily by stellar mass, as more massive stars burn hotter and exhaust their fuel more quickly, while lower mass stars may persist for tens of billions of years. The energy released by fusion provides the outward pressure that balances the inward pull of gravity, and it is the equilibrium between these two forces that governs a star's evolution throughout its life.

Once hydrogen in the core is depleted, the star's evolutionary path depends critically on its mass. The core contracts and heats up while the outer layers expand, transforming the star into a red giant. In lower mass stars, helium burning may be ignited in the expanded core, and the following burning stages produce progressively heavier elements such as carbon and oxygen, before nuclear burning halts and the outer envelope is shed. In more massive stars, the core continues to contract and heat sufficiently to ignite a sequence of burning stages, fusing carbon, neon, oxygen, and silicon in turn, until the core consists mostly of iron. Since iron fusion is endothermic, no further energy can be extracted from the core, and the star becomes unstable to gravitational collapse. This collapse proceeds on a dynamical timescale and results in a supernova, one of the most energetic events in the universe, during which the outer layers of the star are expelled and the core is left behind as a compact remnant.

For these stars, different stages can therefore be identified. First, the initial collapse of dust clouds under their own gravity. Second, the main lifetime during which most stars burn their hydrogen on the main sequence. And finally, the death of stars as they leave the main sequence, during which supernovae may occur or black holes may be formed. Many of the still open questions in modern science arise from these final stages of stellar evolution. For this thesis, however, the main point of interest is the remnants left behind after all these processes, the stellar remnants known as compact objects, which include white dwarfs, neutron stars, and black holes.

1.1 Compact Objects and Binaries

As stated above, compact objects comprise three different categories, based on the progenitor star mass, defined as the stellar mass prior to the supernova event. In order of increasing mass, the first class is that of the white dwarfs, referred to as WDs hereafter. For clarity,

this discussion will focus on stars with initial masses higher than $0.9M_\odot$, as stars below this threshold either fail to ignite entirely, known as brown dwarfs, or have main sequence lifetimes exceeding the current age of the universe.[1] Stars with initial masses between 0.9 and $8M_\odot$ are not sufficiently massive, and therefore not dense or hot enough in their cores, to ignite elements heavier than carbon. As a result, nuclear burning comes to a halt after the star has evolved through the red giant phase and shed its hydrogen envelope through stellar winds. The remaining hydrogen-poor core then collapses under its own gravity, however the electron degeneracy pressure is sufficient to resist the gravitational collapse, leaving behind a WD. The mass of the resulting WD is necessarily below $1.44M_\odot$, known as the Chandrasekhar mass, as any mass exceeding this limit would generate core pressures sufficient to reignite nuclear burning. Should a WD accrete enough mass to surpass the Chandrasekhar limit, a thermonuclear runaway reaction is triggered, giving rise to a type Ia supernova.

Stars with an initial mass between around 8 and $25M_\odot$ are sufficiently massive to progress through all nuclear burning cycles up to iron, at which point burning ceases, as iron fusion is an endothermic process. The star then undergoes collapse, but for this mass range the implosion is halted by neutron degeneracy pressure. The star explodes as a supernova, during which the energy released is sufficient to drive the endothermic photodisintegration reactions, after which protons and electrons in the core combine to form neutrons, leaving behind a stable degenerate neutron core. The infalling outer layers subsequently bounce off this core during the supernova and are expelled into the surrounding space. The masses of neutron stars, referred to as NSs hereafter, are less well constrained than those of WDs, but are generally thought to lie below 2–$3M_\odot$ [Burgio et al., 2021, Tang et al., 2026], above which the neutron degeneracy pressure is no longer sufficient to halt the collapse.

Above an initial mass of around $25M_\odot$, though the precise value of this threshold remains highly uncertain, the gravitational collapse can no longer be halted by any known process, resulting in the formation of a point of infinite density, to which the star collapses during the supernova once its progenitor has exhausted all available nuclear fuel. A black hole, referred to as a BH hereafter, is fully characterised by three quantities: its spin, its mass (and with it the associated radius of the event horizon), and its charge, though in this thesis only the mass will be considered. The evolutionary processes leading up to this end stage are, however, considerably less well understood than those governing WD formation, and remain an active area of research, which will be explored in greater detail in later chapters of this thesis.

Compact objects do not always originate from isolated single stars, but may instead form in binary or even triple star systems [Moe and Di Stefano, 2017]. When stars form from a collapsing dust cloud, a single star may not be sufficient to conserve the total angular momentum of the system, as an excessive rotational momentum would cause the protostar to be disrupted by tidal forces before it can properly form. A binary system then presents itself as a viable alternative outcome. This occurs more frequently for more massive stars, as the dust clouds from which they form are correspondingly larger, increasing the likelihood of binary formation. The precise distribution and composition of binary and triple systems throughout the universe nonetheless remains an active topic of research.

[1] As via $T \propto M^{2/3}$ [Carroll and Ostlie, 2007].

1.2 Introductory Theory

When two compact objects orbit one another in a binary system, the system can be intuitively understood by analogy with two spherical objects rotating in water, generating waves in the surrounding medium. In this analogy, these waves correspond to gravitational waves, or gravitational radiation, which carry away the orbital energy of the binary system. To describe these gravitational waves, a relativistic treatment of the emitted radiation and a prescription of the orbital energy of the source is required.

1.2.1 Gravitational Waves

Following Einstein's theory of general relativity, a notable result is the existence of propagating waves within the fabric of spacetime. To derive this, we begin with Einstein's field equations, in which the gravitational constant G and the speed of light c are set to 1 [2], giving

$$G_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 8\pi T_{\mu\nu}. \tag{1.1}$$

Here $G_{\mu\nu}$ is the Einstein tensor, $R_{\mu\nu}$ and R are the Ricci tensor and scalar respectively, $g_{\mu\nu}$ is the metric tensor, and $T_{\mu\nu}$ is the stress-energy tensor. The construction of these quantities, as well as the full derivation of the equations in this section and the interpretation of the results, can be found in Appendix A.

Gravitational wave solutions to Eq. 1.1 are obtained by assuming vacuum conditions, i.e. setting the stress-energy tensor to zero, and treating spacetime as a small perturbation about flat Minkowski space, such that

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \tag{1.2}$$

where $\eta_{\mu\nu}$ is the Minkowski metric for flat spacetime and $h_{\mu\nu}$ is the perturbation, which is the quantity of interest. Since this is treated as a perturbation, its absolute value must satisfy $|h_{\mu\nu}| \ll 1$. This linearisation serves as an ansatz for Eq. 1.1, and it is the resulting linearised field equations that we seek to solve.

After the derivation presented in Appendix A [Maggiore, 2007, Ellis, 2017, Bertschinger, 2000], the main result is expressed in the transverse-traceless gauge as

$$h^{TT}_{\mu\nu} = C_{\mu\nu}\, e^{ik_\alpha x^\alpha}, \tag{1.3}$$

where k^μ is the wave vector propagating in the z-direction at the speed of light with frequency ω, given by $k^\mu = (\omega, 0, 0, \omega)$, and $C_{\mu\nu}$ is the polarisation tensor

$$C_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & h_+ & h_\times & 0 \\ 0 & h_\times & -h_+ & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \tag{1.4}$$

[2] G and c are only set to 1 for the field equations, they will reappear later in this thesis.

The components h_+ and $h_\times$ can be interpreted in terms of their effect on spacetime through the response of a set of test masses, as illustrated in Figure 1.

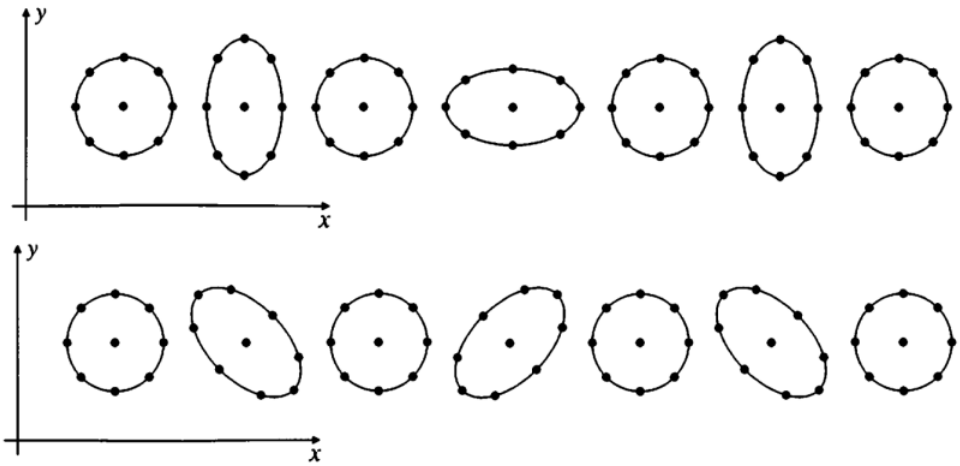


Figure 1: The effect of a passing gravitational wave on a ring of test masses for the two polarisation states, h_+ (top) and $h_\times$ (bottom), over one full wave cycle. The alternating stretching and squeezing of spacetime along perpendicular axes, rotated by 45° between the two polarisations, constitutes the measurable strain $h(t)$ defined in Eq. 1.20 [Carroll, 2019].

1.2.2 Binary systems

Having established a description of gravitational waves, we now turn to the energy source of interest, namely coalescing compact binaries.

For a binary system with component masses m_1 and m_2, we define the separation vector $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$, the total mass $M = m_1 + m_2$, and the reduced mass $\mu = \frac{m_1 m_2}{m_1 + m_2}$. The equations of motion for each body then yield an equation for the relative separation,

$$\begin{aligned} m_1 \ddot{\mathbf{r}}_1 &= -\frac{G m_1 m_2}{r^3}\mathbf{r}, \qquad & m_2 \ddot{\mathbf{r}}_2 &= \frac{G m_1 m_2}{r^3}\mathbf{r}, \\ \ddot{\mathbf{r}} &= -\frac{G(m_1 + m_2)}{r^3}\mathbf{r}, \qquad & \mu \ddot{\mathbf{r}} &= -\frac{G m_1 m_2}{r^3}\mathbf{r}. \end{aligned} \tag{1.5}$$

For a circular orbit with velocity $v = \omega R$, an assumption to which we will return at a later stage, balancing the centripetal acceleration against the gravitational acceleration from Eq. 1.5 gives

$$\frac{v^2}{R} = \frac{GM}{R^2} \implies \omega^2 = \frac{GM}{R^3}, \tag{1.6}$$

from which Kepler's third law follows,

$$a = \left(\frac{GM}{\omega^2}\right)^{1/3}, \tag{1.7}$$

where the orbital separation a is taken as the radius R in the binary system. The gravitational wave frequency ω_{gw} is twice the orbital frequency ω_s, since for a system in circular motion the mass configuration repeats after a rotation of 180°, completing one full gravitational wave cycle in half an orbital period. The total orbital energy is then given by the sum of the kinetic and potential energy, making use of Eqs. 1.6 and 1.7,

$$\begin{aligned} E_{\text{orbit}} = K + U &= \frac{1}{2}\mu\frac{GM}{a} + \left(-\frac{Gm_1m_2}{a}\right) = -\frac{Gm_1m_2}{2a} \\ &= -\left(\frac{\pi^2 G^2 M_c^5 \omega_s^2}{8}\right)^{1/3} \\ &= -\left(\frac{G^2 M_c^5 \omega_{gw}^2}{32}\right)^{1/3}, \end{aligned} \tag{1.8}$$

where the chirp mass is defined as

$$M_c = \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}}. \tag{1.9}$$

Now we want to obtain the power radiated in gravitational waves, for which linearised field equation is solved with a source term $T_{\mu\nu}$, where the source is assumed to be compact relative to the emitted wavelength, i.e. $\lambda \gg R$. This condition permits a multipole expansion of the radiation field. Upon evaluating this expansion [Einstein, 1918], the monopole term vanishes by conservation of mass-energy, the dipole term vanishes by conservation of linear momentum, and the leading-order contribution to the radiated power is therefore the quadrupole term,

$$P_{\text{quad}} = \frac{G}{5c^5}\langle \dddot{Q}_{ij} \dddot{Q}_{ij} \rangle, \tag{1.10}$$

where the reduced (trace-free) quadrupole moment tensor is defined as

$$Q_{ij} = \int \rho(\mathbf{x}) \left(x_i x_j - \frac{1}{3}\delta_{ij} r^2\right) d^3x, \tag{1.11}$$

which encapsulates the shape and orientation of the mass distribution. Evaluated explicitly for a binary source [Peters and Mathews, 1963], this yields

$$P_{\text{quad}} = \frac{32}{5}\frac{G^4}{c^5}\frac{(m_1 m_2)^2 (m_1 + m_2)}{R_a^5}, \tag{1.12}$$

which, upon applying Kepler's third law and expressing the result in terms of ω_{gw}, becomes

$$P_{\text{quad}} = \frac{32}{5}\frac{c^5}{G}\left(\frac{G M_c \omega_{\text{gw}}}{2c^3}\right)^{10/3}. \tag{1.13}$$

Since this power is drawn from the orbital energy of the binary given in Eq. 1.8, equating $-\dot{E}_{\rm orb} = P_{\rm quad}$ yields the frequency evolution

$$\dot{\omega}_{\rm gw} = \frac{96}{5}\pi^{8/3}\left(\frac{GM_c}{c^3}\right)^{5/3}\omega_{\rm gw}^{11/3}, \tag{1.14}$$

$$\dot{\nu} = \frac{96}{5}(2\pi)^{8/3}\left(\frac{GM_c}{c^3}\right)^{5/3}\nu^{11/3}, \tag{1.15}$$

where in the second expression the result has been converted to a frequency in Hz, and the source frequency ν is introduced for convenience in subsequent derivations. Defining the prefactor as K and solving the resulting differential equation yields the time evolution of the orbital frequency,

$$\nu(t)^{-8/3} - \nu_0^{-8/3} = \frac{8}{3}K(t_0 - t). \tag{1.16}$$

1.3 Detecting Gravitational Waves

For the detection of gravitational waves, the motion of at least two masses through spacetime must be measured. These masses follow geodesics governed by

$$\frac{d^2x^i}{d\tau^2} = -\Gamma^i_{\alpha\beta}\frac{dx^\alpha}{d\tau}\frac{dx^\beta}{d\tau}. \tag{1.17}$$

Assuming a +-polarised wave propagating in the z-direction, with one test mass located at the origin and another at x-coordinate L_*, both with four-velocity $u^\alpha_{\rm A} = u^\alpha_{\rm B} = (1,0,0,0)$, the linearised dynamics in the x-direction are captured by substituting into Eq. 1.17,

$$\delta\Gamma^i_{tt} = -\frac{1}{2}\delta^{ij}(\partial_t h_{jt} + \partial_t h_{jt} - \partial_j h_{tt}) = 0, \tag{1.18}$$

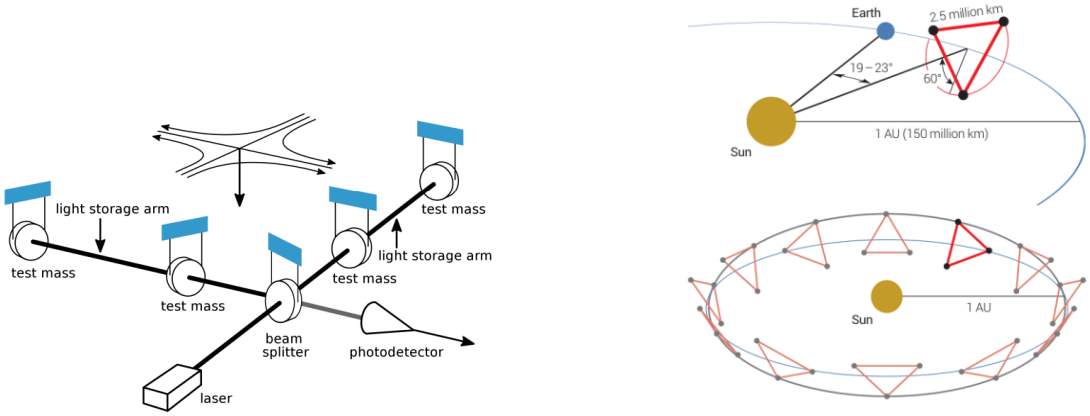
where the linearised Christoffel symbol is taken as per Eq. A.2 in Appendix A. This result implies that the coordinates themselves do not change. The proper length ℓ_x along the x-axis between these coordinates then evolves as

$$\ell_x = \int_0^L \sqrt{g_{xx}}\,dx \approx L_*\left(1 + \frac{1}{2}h_+\right), \tag{1.19}$$

upon substituting the metric definitions from Eqs. 1.2 and 1.4. Repeating this procedure along the y-axis yields a comparison length, from which the gravitational wave strain is defined as

$$h(t) \equiv \frac{\Delta L_*}{L_*} = \frac{\delta\ell_x - \delta\ell_y}{L_*} = h_+(t), \tag{1.20}$$

which constitutes the directly measurable quantity. Physical detectors such as LIGO, VIRGO, and KAGRA, collectively forming the LVK collaboration, measure this strain via laser interferometry, as illustrated in Figure 2.



(a) Schematic of a ground-based laser interferometer, where the interference pattern at the photodetector encodes $h(t)$ [Mayerson et al., 2019].

(b) Orbital configuration of LISA, with arm lengths of 2.5×10^6 km trailing Earth [Amaro-Seoane et al., 2017a].

Figure 2: Gravitational wave detectors. Ground-based detectors such as LIGO, VIRGO and KAGRA use laser interferometry across kilometre-scale arms, while LISA extends this principle into space to achieve sensitivity at much lower frequencies.

In this setup, two laser beams directed along the x- and y-arms interfere destructively in the absence of a gravitational wave. When a gravitational wave passes, the proper length of one arm is slightly altered, causing the beams to interfere constructively whenever the path length difference between the two arms is an integer multiple of the laser wavelength.

The Laser Interferometer Space Antenna (LISA) extends this interferometric principle from the ground into space, employing three spacecrafts to achieve arm lengths far exceeding those of ground-based detectors, thereby enabling sensitivity to gravitational waves at much lower frequencies, as illustrated in Figure 2. For a detailed discussion of the associated engineering challenges, see [LISA Consortium et al., 2024].

1.4 Aims

This thesis applies the theoretical framework developed by [Staelens and Nelemans, 2024] and further elaborated by [Hofman and Nelemans, 2024], which describes the gravitational wave background arising from compact binary systems and is worked out in detail in Chapter 2, to black hole binary systems. While the original work was developed primarily for white dwarf binaries within the frequency band accessible to LISA, the same formalism can be applied to black hole binaries across a broader frequency regime. In Chapters 2 and 3, the principal factors governing this background are examined, with the aim of quantifying their relative contributions when applied to black hole systems. Furthermore, the results obtained within the LISA band are compared to those for white dwarf binaries, in order to provide a more complete characterisation of the astrophysical gravitational wave background in this frequency range.

2 The Astrophysical Gravitational Wave Background

A variety of astrophysical processes give rise to gravitational radiation, with binary systems being of particular interest in this thesis. The superposition of all such processes, integrated over all redshifts, produces a stochastic background that is in principle measurable from Earth. The spectral structure of this astrophysical gravitational wave background, hereafter referred to as the AGWB, encodes information about the rates and physical conditions under which these processes occur, such as the binary merger rate as a function of redshift.

2.1 Theoretical Derivation

To derive an expression for the AGWB, we want to expand the previously established description of a single binary system to a cosmological setting by integrating over the entire population of binaries throughout the universe, after which the relevant frequency regimes and subcases will be identified.

2.1.1 Stochastic Background

Rather than following the evolution of a single binary, we now instead consider the energy emitted within each infinitesimal frequency interval along the evolutionary track of Eq. 1.15, assuming that each such interval is continuously occupied by a binary at that frequency. As one binary evolves out of a given bin, another instantaneously takes its place; the edge cases of binary formation and merger will be addressed in Section 2.1.2. Since we assume that all bins are populated by systems distributed throughout the universe, a normalisation is required. Following convention, we normalise to the critical density of a flat universe, yielding the dimensionless energy density spectrum

$$\Omega_{\rm GW}(f_r) = \frac{1}{\rho_c c^2}\frac{d\varepsilon_{\rm GW}}{d\ln f_r}, \tag{2.1}$$

where $\rho_c = \frac{3H_0^2}{8\pi G}$ is the critical density and f_r is the received gravitational wave frequency. The logarithmic frequency interval $d\ln f_r$ is adopted here as it is the natural measure for a spectrum that spans many orders of magnitude in frequency. The received energy density $\varepsilon_{\rm GW}$ is obtained by integrating the source energy spectrum over all redshifts, weighted by the number density of emitters $N(z)$,

$$\varepsilon_{\rm GW} = \int_0^\infty \frac{df_r}{f_r} f_r \int_0^\infty dz\, N(z)\frac{dE_{\rm GW}}{df_e}, \tag{2.2}$$

where the cosmological redshift between the emitted frequency f_e and the received frequency f_r is accounted for implicitly through $f_e = f_r(1+z)$. Substituting into the definition of $\Omega_{\rm GW}$,

the $d\ln f_r$ term cancels cleanly, yielding

$$\Omega_{\rm GW}(f_r) = \frac{f_r}{\rho_c c^2}\int_0^\infty dz\, N(z)\frac{dE_{\rm GW}}{df_e}. \tag{2.3}$$

Even in the absence of any specific population model, a general scaling relation for the dimensionless energy spectrum can be inferred. Taking the derivative of the orbital energy in Eq. 1.8 as a crude ansatz, the characteristic $\Omega_{\rm GW} \propto f_r^{2/3}$ dependence emerges naturally, neglecting edge cases near formation and merger.

The right-hand side of Eq. 2.3 can be encapsulated by the specific flux $\mathcal{F}_{f_r}$ [Farmer and Phinney, 2003] [3], defined as the specific luminosity ℓ_{f_e} of all sources integrated over cosmic time, properly redshifted to the observer frame,

$$\mathcal{F}_{f_r} = \int_{z=0}^\infty \frac{\ell_{f_e}(z)}{(1+z)^2}\left(\frac{df_e}{df_r}\right) d\chi(z), \tag{2.4}$$

where ℓ_{f_e} is the specific luminosity of a source at emitted frequency f_e, the factor $(1+z)^{-2}$ accounts for the redshifting of energy and the time dilation of the arrival rate, and $d\chi(z)$ is the comoving volume element. The dimensionless energy density spectrum then takes the compact form

$$\Omega(f_r) = \frac{1}{\rho_c c^3} f_r \mathcal{F}_{f_r} = \frac{1}{\rho_c c^3}\frac{f_r \mathcal{F}_{f_{r_1}\to f_{r_2}}}{f_{r_2}-f_{r_1}}. \tag{2.5}$$

To further evaluate Eq. 2.4, we consider the specific flux received within a frequency bin with edges f_{r_1} and f_{r_2}, and substitute the definition of the specific luminosity. The luminosity density is built up from the number density $n_k(\nu, z)$ of binaries of type k at orbital frequency ν and redshift z, each contributing a single-binary luminosity $L_k(\nu)$. The connection between the orbital frequency ν and the emitted gravitational wave frequency f_e is enforced by the delta function $\delta(f_e - 2\nu)$, reflecting the earlier established relation $f_e = 2\nu$. The bin edges are expressed in terms of the emitted frequency via $f_e = f_r(1+z)$, accounting for the cosmological redshift between the emitted and received frames. Summing over all binary types k and integrating over all redshifts then gives

$$\begin{aligned}\mathcal{F}_{f_{r_1}\to f_{r_2}} &= \sum_k \int_{z=0}^\infty \frac{d\chi(z)}{(1+z)^2}\int_{f_{r_1}(1+z)}^{f_{r_2}(1+z)}\int_{\nu_0}^{\nu_{\rm merge}} n_k(\nu,z)L_k(\nu)\delta(f_e-2\nu)\,df_e\,d\nu \\ &= \sum_k \int_{z=0}^\infty \frac{d\chi(z)}{(1+z)^2}\int_{\nu_{\rm min}}^{\nu_{\rm max}} n_k(\nu,z)L_k(\nu)\,d\nu. \end{aligned} \tag{2.6}$$

Here in the second step weve used the property of the delta function and we have now switched to $\nu_{\rm min} = \max\{\nu_0,\, f_{r_1}(1+z)/2\}, \nu_{\rm max} = \min\{\nu_{\rm merge},\, f_{r_2}(1+z)/2\}$. to account for the computation of the edge cases to which we will come back to in Section 2.1.2.

Now to compute the frequency integral we need to evaluate n and L, from which the latter of the two can be defined by Eq. 1.13. The number of systems per bin $n_{\rm bin}$ is filled according

[3] In [Farmer and Phinney, 2003] the derivation is done over time, while here we use redshift z, leading to the extra factor of $(1+z)^{-1}$

to the relation established in Eq. 1.14, but still requires normalisation by a factor A. This factor is determined by demanding that the integral of the number density over the bin equals the normalised number of stars formed in that bin,

$$\int_{\nu_{\min}}^{\nu_{\max}} A\nu^{-11/3}\, d\nu = n_{\text{bin}} = \frac{\psi(z+z^*)\cdot\Delta t(k;\text{bin})}{M_{\text{SF}}}, \tag{2.7}$$

where $\psi(z+z^*)$ is the star formation rate evaluated at the time corresponding to redshift $z+z^*$, with z^* encoding the delay between binary formation and the time the system enters the frequency bin. The time spent traversing the bin, $\Delta t(k;\text{bin})$, is given by Eq. 1.16, and M_{SF} is the initial stellar mass of the population synthesis used to normalise the number density. Solving for A gives

$$A = \frac{8}{3}\frac{1}{\nu_{\min}^{-8/3} - \nu_{\max}^{-8/3}}\frac{\psi(z+z^*(\Delta t))}{M_{\text{SF}}}\cdot\Delta t(k;\text{bin}). \tag{2.8}$$

Substituting $n_k = A\nu^{-11/3}$ and L_k from Eq. 1.13 into the frequency integral, the integrand reduces to $\nu^{-1/3}$. Collecting all numerical prefactors then yields

$$\mathcal{F}_{f_{r_1}\to f_{r_2}} = 8.8{\cdot}10^{38}\sum_k\left(\frac{M_k}{M_\odot}\right)^{10/3}\int_{z=0}^{\infty}\frac{\nu_{\max}^{2/3}-\nu_{\min}^{2/3}}{\nu_{\min}^{-8/3}-\nu_{\max}^{-8/3}}\frac{\psi(z;k;\text{bin})}{M_{\text{SF}}}\Delta t(k;\text{bin})\frac{d\chi(z)}{(1+z)^2}\,\frac{\text{J}\,\text{Hz}^{-10/3}}{\text{s}}. \tag{2.9}$$

This provides all the ingredients necessary to evaluate Ω_{GW}. The remaining subtlety concerns the evaluation of ψ at $z+z^*$. For each binary k, we determine z^* such that it corresponds to the cosmic time $t_{k,0}+\tau_k$, where $t_{k,0}$ is the formation time of the binary as given by the population synthesis, and τ_k is the delay time computed from Eq. 1.16 with ν_0 taken from the population synthesis and the bin edge converted to the emitted frame via $\nu_{\min} = f_{r_1}(1+z)/2$.

2.1.2 Regimes and Cases

Before we compute the background, we need to distinguish between three cases and two regimes, which were alluded to earlier. The connection between frequency evolution and time evolution gives rise to cases in which a binary evolves so slowly that it does not leave a frequency bin within the Hubble time. We refer to this as the static regime, since on our timescale the binaries within it appear stationary. Using Eq. 1.16, we can substitute t_H for the time, take $f(t)$ to infinity, and solve for $f(0)$ to obtain an approximation for $f_{\min}$. Since K is chirp mass dependent, this approximation can be evaluated for all compact binary types, as shown in Table 1. Whereas the inspiral regime followed a $\Omega \propto f^{2/3}$ dependence, the static regime behaves differently. Since there is approximately no frequency evolution, dE/df is determined by the energy of a static system, that is its luminosity from Eq. 1.13 multiplied by time, weighted by the number of systems per frequency bin. Assuming a uniform distribution across $\log f$ [Farmer and Phinney, 2003], this yields $\Omega \propto f^{10/3}$.

To establish a maximum frequency we need to identify a minimum separation at which the inspiral calculation is terminated.[4] A reasonable assumption for this termination point is the innermost stable circular orbit (ISCO), the smallest orbital radius at which a test mass can maintain a stable circular orbit around a massive body. Inside this radius no stable orbit exists and the object inevitably plunges inward, the adiabatic inspiral approximation breaks down, and post-Newtonian gravity ceases to be valid. Numerical relativity is required to follow the subsequent merger and ringdown. We therefore take the ISCO as the natural endpoint of our inspiral calculation. To locate this endpoint, we switch from the Keplerian orbital frequency to a relativistic, gauge-invariant frequency parameter, defined as

$$x_0 \equiv \left(\frac{GM\omega_0}{c^3}\right)^{2/3}, \tag{2.10}$$

Rather than a simple velocity ratio, x_0 tracks the relativistic evolution of the orbit in a coordinate-independent way, reducing to the Newtonian result in the weak-field limit. The ISCO is then reached when x_0 attains its ISCO value $x_{\rm ISCO}$, beyond which the system transitions to merger. To also capture the mass-ratio dependence correctly, which cancels at 1PN (Post-Newtonian) but contributes at higher order, we adopt the 2PN result of [Blanchet and Iyer, 2003],

$$x_{\rm ISCO}^{\rm 2PN} = \frac{3}{14\nu}\left(1 - \sqrt{1 - \frac{14\nu}{9}}\right), \qquad \nu = \frac{m_1 m_2}{(m_1+m_2)^2}, \tag{2.11}$$

where ν is the symmetric mass ratio. Substituting $x_{\rm ISCO}$ into the relativistic Keplerian relation and solving for the corresponding gravitational wave frequency then yields the maximum frequency,

$$f_{\rm max} = \frac{c^3}{2\pi GM}\left(x_{\rm ISCO}\right)^{3/2}. \tag{2.12}$$

Eq. 2.12 can again be used as an approximation with the result in Table 1 for the black hole case, while the $f_{\rm max}$ for WD and NS uses Roche Lobe contact like [Staelens and Nelemans, 2024], with using the Eggleton formula [Eggleton, 1983], which isnt any further relevant for this thesis.

System	$M_1 = M_2$ $[M_\odot]$	M_c $[M_\odot]$	$f_{\rm min}$ [Hz]	$f_{\rm max}$ [Hz]
WD	0.3	0.26	4.3×10^{-5}	3.6×10^{-2}
NS	1.4	1.2	1.6×10^{-5}	1.4×10^{3}
smBH	10	8.7	4.7×10^{-6}	2.6×10^{2}
SMBH	10^6	8.7×10^5	3.6×10^{-9}	2.6×10^{-3}

Table 1: Approximate minimum and maximum gravitational wave frequencies for equal-mass compact binary systems. $f_{\rm min}$ is computed from the static regime approximation, $f_{\rm max}$ for WDs and NSs from the Eggleton Roche lobe overflow condition [Eggleton, 1983], and for BHs from the 2PN ISCO condition [Blanchet and Iyer, 2003].

[4]Note that this is not the real $f_{\rm max}$ of the binary, and that more complete models including merger and ringdown will exhibit a higher peak at a higher frequency.

Within the frequency range defined by $f_{\min}$ and $f_{\max}$, three cases can be distinguished for a given frequency bin: the generic case, the birth case, and the merger case. In the generic case, the binary evolves through the entire bin, whereas in the birth and merger cases the binary respectively enters or exits the bin at some point within it, requiring special treatment of the integration limits.

Generic case. Since in the generic case the binary traverses the entire bin, the bin edges can be used directly as integration limits, with $\nu_{\min} = f_{r_1}(1+z)/2$ and $\nu_{\max} = f_{r_2}(1+z)/2$, where the factor of $(1+z)/2$ converts the received frequency bin edges to emitted orbital frequencies. Substituting these into Eq. 2.9, the $\nu_{\min,\max}$ dependence simplifies analytically, and converting to SI units and discretising over redshift for computational purposes, yields

$$\begin{aligned}\mathcal{F}_{f_{r_1}\to f_{r_2}} &= 1.89\cdot 10^{-9}\left(f_{r_2}^{2/3} - f_{r_1}^{2/3}\right) \\ &\quad\times \sum_{k,i}\left(\frac{M_k}{M_\odot}\right)^{5/3}\frac{\psi(k;\text{bin};z_i)}{M_{\rm SF}/M_\odot}\cdot(1+z_i)^{-4/3}\cdot\frac{\Delta\chi(z_i)}{\text{Mpc}} \quad [\text{kg s}^{-3}].\end{aligned} \tag{2.13}$$

Substituting into Eq. 2.3 then gives the final result for the generic case,

$$\Omega = 8.10\cdot 10^{-9}\cdot f_r\frac{\left(f_{r_2}^{2/3} - f_{r_1}^{2/3}\right)}{(f_{r_2} - f_{r_1})}\sum_{k,i}\left(\frac{M_k}{M_\odot}\right)^{5/3}\cdot\frac{\psi(z_i + z^*_{k,\text{bin}})}{M_{\rm SF}/M_\odot}\cdot(1+z_i)^{-4/3}\cdot\frac{\Delta\chi(z_i)}{\text{Mpc}}. \tag{2.14}$$

Birth and merger cases. For the birth and merger cases the same procedure is followed, but $\nu_{\min}$ and $\nu_{\max}$ can no longer be replaced by the bin edges, one of the limits is instead set by ν_0 or $\nu_{\rm merge}$ (which we will set with $f_{\max}$), whichever falls within the bin. Since the full substitution $\nu_{\min,\max} = f_r(1+z)/2$ can no longer be performed, $\nu_{\min}$ and $\nu_{\max}$ must be retained explicitly, and the $(1+z)$ factors no longer simplify. The resulting Ω then takes the form

$$\Omega = 1.28\cdot 10^{-8}\cdot f_r\sum_{k,i}\frac{\nu_{\max}^{2/3} - \nu_{\min}^{2/3}}{f_{r_2} - f_{r_1}}\left(\frac{M_k}{M_\odot}\right)^{5/3}\frac{\psi(k;\text{bin};z_i)}{M_{\rm SF}/M_\odot}(1+z_i)^{-2}\cdot\frac{\Delta\chi(z_i)}{\text{Mpc}}. \tag{2.15}$$

Together, Eq. 1.16 for $\psi(t_k+\tau)$ and Eqs. 2.14 and 2.15 fully determine the total AGWB, which takes m_1, m_2, ν_0 and t_0 of every binary k as input, to be retrieved from the population synthesis.

2.2 Binary Evolution Parameters

Having established a method to compute the AGWB, we now investigate the parameters that influence the result. The merger rate R describes the rate at which BH binaries (BHB) form in the population synthesis, encapsulating all the underlying physical processes, and its influence on Ω can be understood through the behaviour of the number density $N(f)$, which we derive here.

To do so, we reverse the time evolution of the binaries and consider systems created at a birth rate R, all forming at the same orbital frequency ν_m at which they would normally

merge, subsequently evolving towards lower frequencies according to Eq. 1.15. The time for a system to evolve from ν_m down to frequency ν_0 is $t_m(\nu_0)$, meaning there are $t_m(\nu_0) \cdot R$ systems between ν_0 and ν_m at any given time. Taking the derivative with respect to frequency then gives the number density at a given frequency f,

$$N(f) \propto \frac{R\, t_m(f)}{f} \propto R\, M^{-5/3} f^{-11/3}. \tag{2.16}$$

Substituting this into Eqs. 2.14 and 2.15 alongside the single-binary luminosity L from Eq. 1.13, the scaling of Ω with the merger rate follows as

$$\Omega \propto N(f)\, L\, f \propto R\, M^{5/3} f^{2/3}, \tag{2.17}$$

demonstrating the direct linear dependence of the AGWB on the merger rate. The ingredients that thus impact the height of the resulting curve are the chirp mass, together with the parameters that vary the merger rate, such as orbital separation and time of merger, which in turn depend on the different processes governing binary evolution.

2.2.1 Mass Transfer

The population synthesis code used in this thesis is SeBa [Nelemans et al., 2001, Portegies Zwart and Verbunt, 1996, Portegies Zwart and Yungelson, 1998, Toonen et al., 2012], which evolves a large number of binary systems, from the zero-age main sequence through all subsequent stages of stellar and binary evolution up to and including the compact remnant phase. The code employs fast analytical recipes and accounts for a wide range of physical processes, including mass transfer and common envelope evolution, which will be discussed below. Different metallicities can be specified as input parameters. The output consists of the end products of all evolved binaries over a Hubble time, which may be a surviving binary system or one or two single compact objects in the case of a merger or disruption. The binaries are initialised as isolated systems, meaning that no tertiary companions or environmental effects such as dynamical interactions in dense stellar clusters are taken into account.

During the evolution of binary systems in SeBa, several physical processes must be accounted for that can significantly influence the final population. For the purposes of gravitational wave background modelling, two main formation channels are relevant: the stable mass transfer channel and the common envelope evolution (CEE) channel, both of which are considered the dominant pathways to compact binary formation [Dominik et al., 2012]. A formation channel describes the evolutionary path two main sequence stars follow to arrive at a given compact binary configuration, and the physical assumptions made along this path directly influence the properties of the resulting population.

Both channels share a common starting point, illustrated in Figure 3, two zero-age main sequence stars (ZAMS) evolve until the more massive primary expands into a (super)giant phase, at which point it overflows its Roche lobe, initiating mass transfer to the secondary. The secondary then also evolves off the main sequence, at which point the system either enters the stable channel or the CEE channel, both resulting in a double compact binary. In the

stable case, mass transfer proceeds gradually and conservatively. Under conservation of total mass and angular momentum, the orbital separation decreases when the donor is the more massive star and widens when it is the less massive one, as mass is redistributed between the two components.

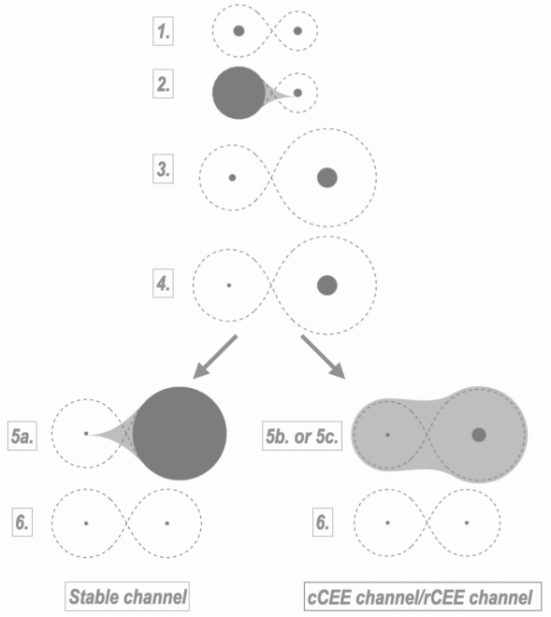


Figure 3: The two main BHB formation channels. Steps 1–4 are common to both: two ZAMS stars evolve until the primary overflows its Roche lobe, initiating mass transfer. The system then either follows the stable channel (5a) or the cCEE/rCEE channel (5b/5c), both resulting in a double compact binary. [Dorozsmai and Toonen, 2024]

In the unstable case, the overflow leads to a common envelope phase, in which both stellar cores are embedded within the envelope of the more extended star. When the envelope is convective the phase is referred to as convective CEE (cCEE), and when radiative as radiative CEE (rCEE). Two main formalisms are used to describe this phase. The first, the α formalism, is based on conservation of energy, in which orbital energy is transferred into the binding energy of the envelope through drag. The binding energy of the envelope is given

$$E_{\rm bind} = \frac{GM_1 M_{\rm env}}{\lambda R_1}, \tag{2.18}$$

where λ is a dimensionless structure parameter encoding the binding properties of the stellar envelope and $M_{\rm env}$ is the envelope mass. The α formalism then equates this to the energy extracted from the orbit,

$$\frac{GM_1 M_{\rm env}}{\lambda R_1} = \alpha \left(\frac{GM_{1,f} M_2}{2a_f} - \frac{GM_1 M_2}{2a_i} \right), \tag{2.19}$$

where α is the efficiency parameter. Since λ is uncertain and often poorly constrained, it is frequently absorbed into α, which in part explains why values of $\alpha > 1$ are physically admissible and can be interpreted as additional energy reservoirs contributing to the envelope ejection [Webbink, 1984]. From the definition of orbital energy, larger α corresponds to a wider post-CEE separation.

The second, the γ formalism, is based on conservation of angular momentum, stating that angular momentum is carried away with the ejected mass via

$$\frac{J_{\text{init}} - J_{\text{final}}}{J_{\text{init}}} = \gamma \frac{\Delta M}{M + m}, \tag{2.20}$$

where γ is the efficiency parameter. With $J \propto M_1 M_2 \sqrt{a}$, the γ formalism predicts wider post-CEE separations than the α formalism for the same mass loss, and was introduced specifically to explain systems observed to have wider separations than the α formalism can account for. It should be noted, however, that the γ formalism was developed specifically for systems involving two extended stars and loses its physical motivation when one component is already a compact object [Nelemans et al., 2000].

When the first mass transfer phase ends with the ejection of the hydrogen envelope, a stripped helium star remains. The subsequent evolution can follow multiple paths, but for a gravitational wave background population the most relevant scenario involves a secondary, and sometimes tertiary, CEE phase. In these later phases only the α formalism is considered, as one component is now a compact object, for which the γ formalism is not physically motivated [Nelemans et al., 2001]. In this work, two values are adopted: $\alpha = 1$, corresponding to perfect efficiency in which all orbital energy goes into ejecting the envelope, and $\alpha = 4$, representing the case where additional energy reservoirs contribute to the ejection.

The formation channel and CEE formalism directly affect the merger rate and hence the AGWB. As established in Eq. 2.17, any process that modifies the number of binaries reaching small enough orbital separations to inspiral within a Hubble time will leave a direct imprint on the background. The choice of α or γ, and the resulting post-CEE orbital separation, therefore constitutes one of the principal sources of uncertainty in the predicted AGWB.

2.2.2 Unvaried Parameters

SeBa makes a number of additional physical assumptions during the binary evolution that will not be varied in this work, but which may have a significant impact on the final population [Dorozsmai and Toonen, 2024, Breivik, 2026]. The most important of these include the Roche lobe overflow behaviour (mass transfer efficiency, angular momentum loss, and stability criteria [Marchant, 2025]), the natal kick velocity to NSs and BHs at formation, the fraction of fallback onto the stellar remnant, the remnant mass prescription, stellar wind mass and angular momentum loss, the circularisation and synchronisation of tidal interactions, the accretion efficiency, and the Eddington accretion limit.

As argued in the previous section, the CEE formalism is the dominant factor determining post-CEE orbital separations, and therefore the merger rate and the overall amplitude of the AGWB. Given that $\tau_{\text{merge}} \propto a^4$, even moderate changes in the assumed α can shift entire

sub-populations across the Hubble time threshold. The remaining parameters listed above introduce additional uncertainty, but their effects are expected to be secondary to that of the CEE formalism for the BHB population considered here.

2.3 Eccentricity

Having discussed the population synthesis parameters that influence the AGWB, we now turn to an assumption made in the theoretical framework itself, that binary orbits can be treated as circular. For an eccentric orbit, [Peters and Mathews, 1963] showed that the total radiated power is modified by an enhancement factor,

$$L_{\rm GW}(M_1, M_2, f_p, e) = L_{\rm GW,circ}(M_1, M_2, f_p)\, F(e), \tag{2.21}$$

$$F(e) \equiv \frac{1 + \frac{73}{24}e^2 + \frac{37}{96}e^4}{(1-e^2)^{7/2}}. \tag{2.22}$$

Furthermore, for an eccentric orbit the argument that the mass configuration repeats every 180° no longer holds, and gravitational wave power is distributed across all integer harmonics of the orbital frequency, with $f_r = n f_p$ for each harmonic n. By differentiating the orbital energy Eq. 1.8 and equating the result to the power loss from Eq. 2.21, the evolution of the semi-major axis is obtained. Combining this with the angular momentum loss in a Keplerian orbit then yields the coupled evolution equations [Maggiore, 2007],

$$\frac{da}{dt} = -\frac{64}{5}\frac{G^3 M_1 M_2 M_{\rm tot}}{c^5 a^3} F(e), \tag{2.23}$$

$$\frac{de}{dt} = -\frac{304}{15}\frac{G^3 M_1 M_2 M_{\rm tot}}{c^5 a^4 (1-e^2)^{5/2}} e \left(1 + \frac{121}{304}e^2\right), \tag{2.24}$$

which show that eccentric orbits simultaneously shrink and circularise. Using these two equations yields a relation between e and f_e that can be integrated to give the eccentricity evolution as a function of frequency [Enoki and Nagashima, 2007],

$$\frac{f_e}{f_{e,0}} = \left\{ \frac{1-e_0^2}{1-e^2} \left(\frac{e}{e_0}\right)^{\frac{12}{19}} \left[\frac{1+\frac{121}{304}e^2}{1+\frac{121}{304}e_0^2}\right]^{\frac{870}{2299}} \right\}^{-3/2}. \tag{2.25}$$

To account for the redistribution of power across harmonics, we define $g(n,e)$ as the fraction of the total radiated power carried by harmonic n, satisfying $\sum_n g(n,e) = F(e)$. Since the Keplerian orbit is periodic but not sinusoidal, its Fourier decomposition naturally produces Bessel function coefficients, giving [Peters and Mathews, 1963, Fitchett, 1983],

$$g(n,e) \equiv \frac{n^4}{32}\Bigg\{ \Big[J_{n-2}(ne) - 2eJ_{n-1}(ne) + \frac{2}{n}J_n(ne) + 2eJ_{n+1}(ne) - J_{n+2}(ne)\Big]^2 + (1-e^2)\left[J_{n-2}(ne) - 2eJ_n(ne) + J_{n+2}(ne)\right]^2 + \frac{4}{3n^2}\left[J_n(ne)\right]^2 \Bigg\}. \tag{2.26}$$

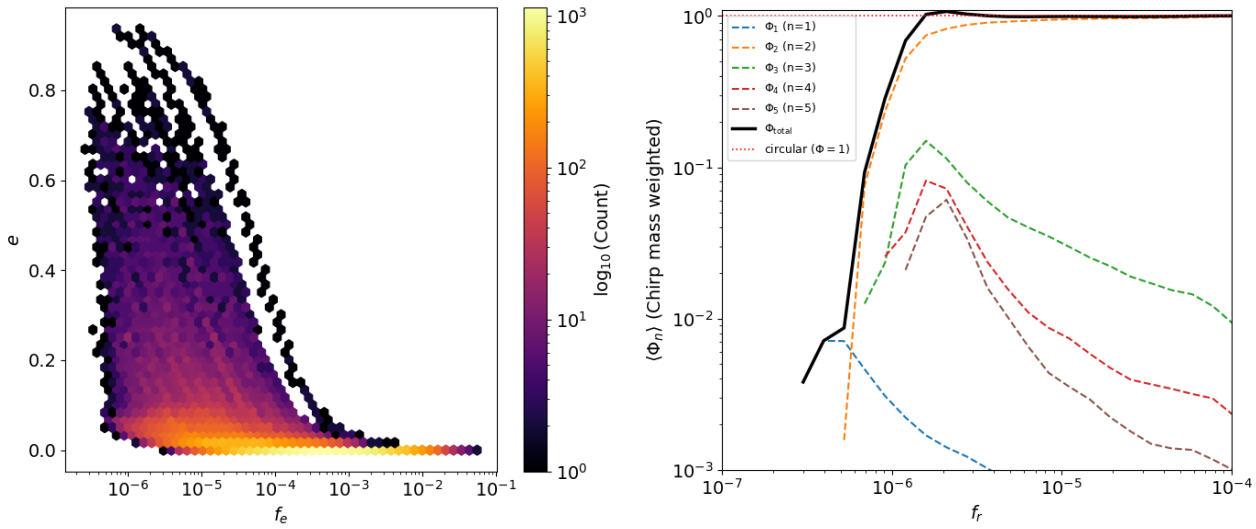
The net correction factor Φ to the circular AGWB is then obtained by comparing the power in each harmonic to the circular case, in which all power goes into $n = 2$. Following [Enoki and Nagashima, 2007], this gives

$$\Phi \equiv \sum_{n=1}^{\infty} \Phi_n, \tag{2.27}$$

$$\Phi_n \equiv \left(\frac{2}{n}\right)^{2/3} \frac{g(n,e)}{F(e)}, \tag{2.28}$$

where the eccentricity e at each frequency is evolved via Eq. 2.25, and the eccentric AGWB is then given by $\Omega_{\rm eccentric}(f_r) = \Phi(f_r) \cdot \Omega_{\rm circular}(f_r)$. To assess whether the circular orbit assumption is valid, a test population of approximately 750 BHB systems was simulated with SeBa.[5] Figure 4a shows the eccentricity distribution as a function of emitted frequency f_e, computed via Eq. 2.25 for each system using its initial $(e_0, f_{e,0})$ from the population synthesis. The concentration of systems near $e \approx 0$ at frequencies above $\sim 10^{-5}$ Hz confirms that the population circularises rapidly, within roughly two orders of magnitude in frequency from the initial $f_{e,0}$. To quantify the impact on the AGWB, Φ is computed per frequency bin with each system weighted by $M_c^{5/3}$, since more massive systems contribute more to the background. The result is shown in Figure 4b, where the contributions of individual harmonics Φ_n are also shown. At low frequencies the higher harmonics carry a non-negligible fraction of the power, but $\Phi_{\rm total}$ goes rapidly to 1 within approximately one order of magnitude above the lowest frequency, confirming that the circular approximation holds across essentially the entire frequency range of interest. This is consistent with the findings of [Perigois et al., 2021], where eccentricity was included explicitly and the AGWB was found to be nearly indistinguishable.

[5]For this specific test, the $\gamma\alpha$ mass transfer prescription with $\alpha = 4$ at metallicity $z = 0.02$ was used. The qualitative conclusion is independent of these choices.



(a) Eccentricity as a function of f_e, with colour indicating system count per bin.

(b) Chirp-mass-weighted $\langle\Phi_n\rangle$ per harmonic and Φ_{total}, with the circular reference $\Phi = 1$ dotted.

Figure 4: Eccentricity correction to the AGWB for a test BHB population.

2.4 Star Formation History and Metallicity

The other factor in the theoretical framework from which uncertainties may arise is the star formation rate ψ. It is thought to be changing over time, and thus redshift, and the main thing done to retrieve this expression is to couple the equations of chemical evolution to the spectroscopic and photo-metric characteristics of the universe. This was done in Madau and Dickinson [2014] and Dickinson to arrive at a crude curve of the star formation rate over time, the star formation history (SFH)

$$\psi(z) = 0.015 \frac{(1+z)^{2.7}}{1 + \left(\frac{1+z}{2.9}\right)^{5.6}}. \tag{2.29}$$

When using this metallicity independent ψ, one can do a test run with the same population from the previous section to find if this is a suitable ψ. For every metallicity preset in SeBa, [Madau and Dickinson, 2014] was used for ψ, and the seven metallicity runs for the AGWB are shown in Figure 5.

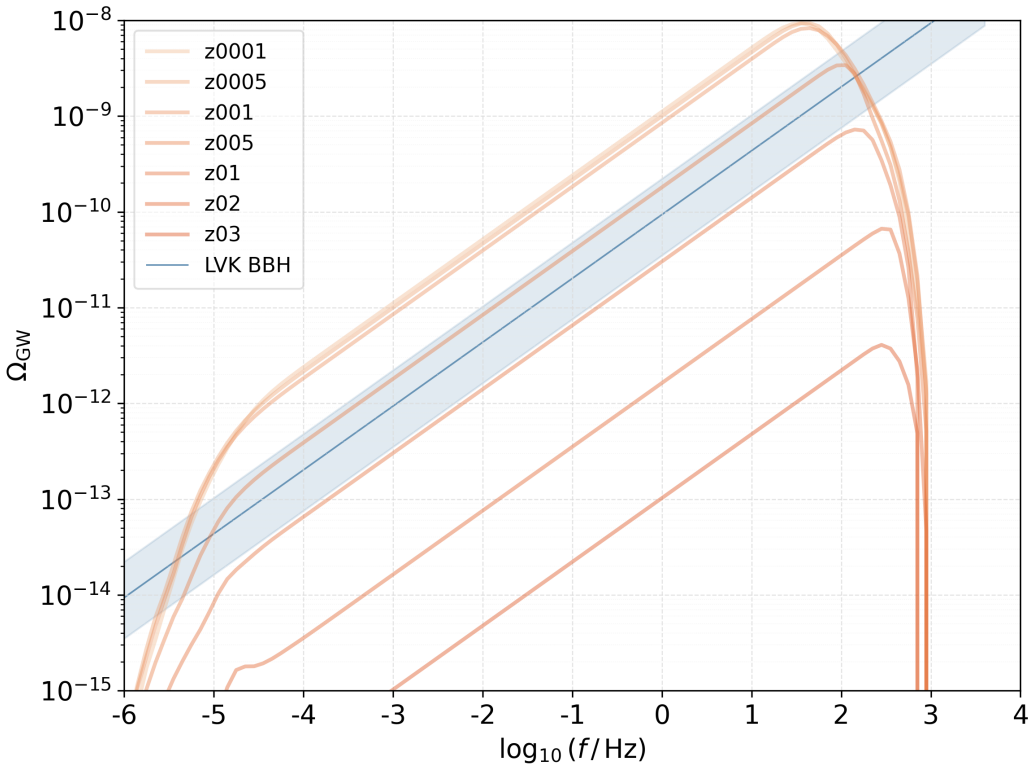


Figure 5: The resulting AGWB for different metallicities using a metallicity independent SFH. The LVK BBH line is a predicted black hole line by measurements of LIGO, VIRGO and KAGRA, which will be more thoroughly discussed in the next section.

As can be readily seen, a metallicity independent ψ results in wildly different amplitudes across the metallicity runs, though this could have been expected for black holes, since metallicity has a strong impact on the merger rate of BHB systems. A higher metallicity star has more metal ions in its envelope, which significantly increases the opacity. This in turn drives stronger stellar winds, stripping more of the envelope and resulting in a lower final mass. Consequently, higher metallicity populations produce fewer and lighter black holes in the binary population [6]. This is clearly reflected in Figure 5, where the higher metallicity curves show a substantially lower amplitude. To combat this problem we must choose a metallicity dependent ψ, for which we will use the same ones as in [Hofman and Nelemans, 2024]. These models are again constructed by coupling chemical evolution to the spectroscopic and photometric properties of observed galaxies at different redshifts, yielding the star formation rate density (SFRD) as a function of both redshift and metallicity. The metallicity ranges from $Z = 0.000005$ to $Z = 0.13$, and the redshift from $z = 0$ to $z = 8$, binned on a logarithmic

[6]This argument does not apply to white dwarfs, whose progenitors are low to intermediate mass stars where radiation-driven, line-opacity-dependent winds are weak. Mass loss in these stars is dominated by pulsation and dust-driven envelope ejection on the AGB, making the final mass and formation rate far less sensitive to metallicity.

scale. The data is binned into six metallicity bins, each assigned to the nearest available SeBa metallicity: $Z \in \{0.0001, 0.001, 0.005, 0.01, 0.02, 0.03\}$, where the upper bin is capped at $Z = 0.03$ due to the limitations of SeBa. Five models are used: LZ19, MZ19 and HZ19 from [Chruślińska and Nelemans, 2019], and LZ21 and HZ21 from [Chruślińska et al., 2021]. The labels refer to the metallicity at which each model forms the most stellar mass, and the 2021 models lie systematically higher in total SFRD due to the inclusion of starburst galaxies. All models peak near $z \approx 1.8$, as seen in Figure 6.

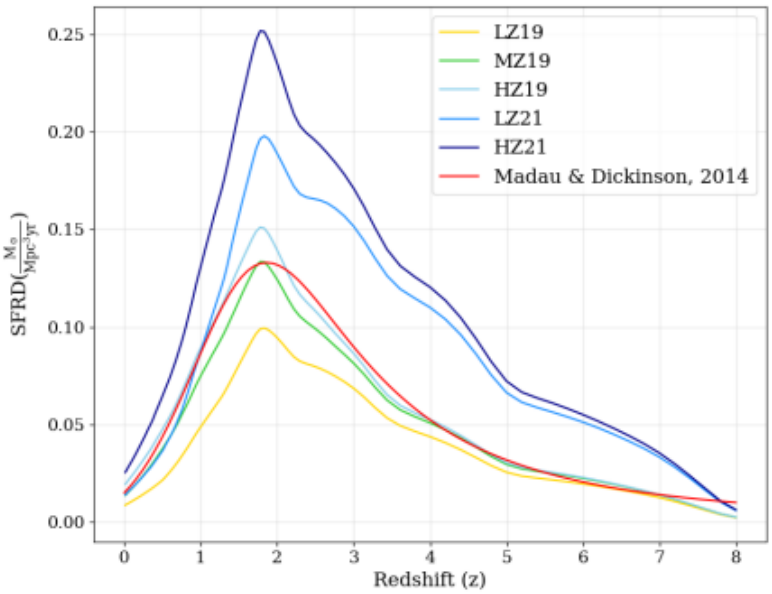


Figure 6: Total SFRD vs redshift for the six different SFHs [Hofman and Nelemans, 2024].

2.5 Previous Estimates

As was already hinted at in the previous section, we want to have some upper limit estimates to get a rough comparison from our simulation data to real observed constraints. The LVK collaboration has searched for a stochastic GWB in the combined data from observing runs O1 through O4a. Upper limits on the GWB amplitude have been placed using a cross-correlation search, in which the data are compared to an expected power-law dependence of the energy density spectrum. The current upper limit at 95% confidence level for a power-law spectrum with $\alpha = 2/3$ is [LIGO Scientific Collaboration et al., 2025]

$$\Omega_{\rm GW}(25\,{\rm Hz}) \leq 2.0 \times 10^{-9}. \tag{2.30}$$

Combining this upper limit with population inference from the GWTC-4 catalog, the predicted contributions to the GWB from coalescing compact binaries are estimated to be

$$\Omega_{\rm BBH}(25\,{\rm Hz}) = 0.8^{+1.1}_{-0.5} \times 10^{-9}, \tag{2.31}$$

$$\Omega_{\rm CBC}(25\,{\rm Hz}) = 0.9^{+1.1}_{-0.5} \times 10^{-9}. \tag{2.32}$$

Ground-based interferometers such as LIGO and Virgo operate in the frequency range $f \sim 10$–10^3 Hz, in which BHB and NS binaries are expected to contribute to the background. These upper limits will be used as a point of comparison when discussing the results of the population synthesis in the following sections.

We will also use a WD curve as a comparison to our BHB results, which was also constructed using the model built upon from [Staelens and Nelemans, 2024] and [Hofman and Nelemans, 2024], the specific settings for that one curve will be shown in the legend of each figure. Detection curves for the ET, LISA and DECIGO telescopes are from [Amaro-Seoane et al., 2017b, Alonso et al., 2020], [Hild et al., 2011] and [Kawamura et al., 2021] respectively.

3 Results

This section consists of two parts. In the first, we compare the total $\Omega_{\rm GW}$ across mass transfer models under a single SFRD, and across SFRD models for each mass transfer variation. The second part examines the population details of the different models to give more insight into the origin of the total results.

3.1 Main Simulation Results

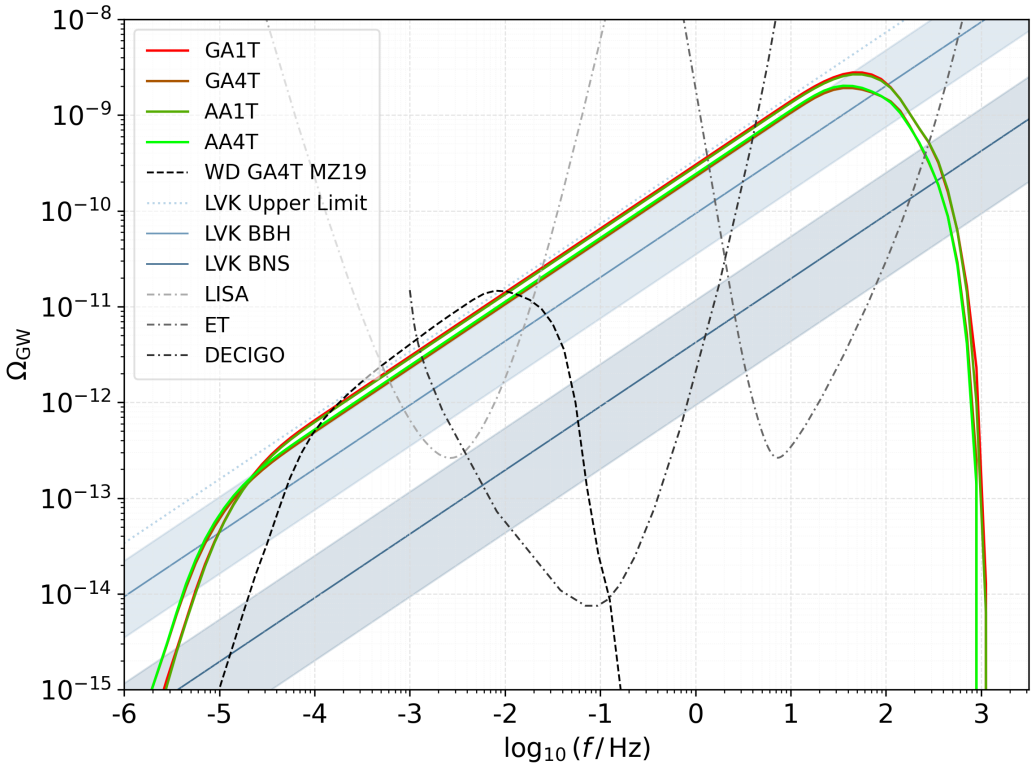
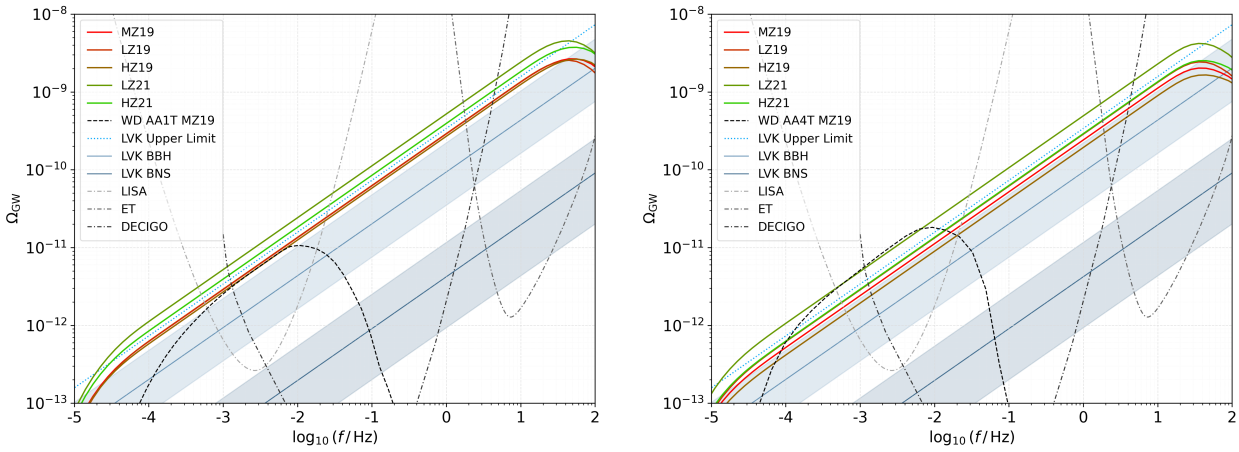


Figure 7: GWB curves for each mass transfer model at SFRD MZ19. G and A stand for γ and α respectively, the number following is the α value, and T indicates time integration. In Appendix B the time and redshift integration comparison can be found, showing no difference in results [Staelens and Nelemans, 2024].

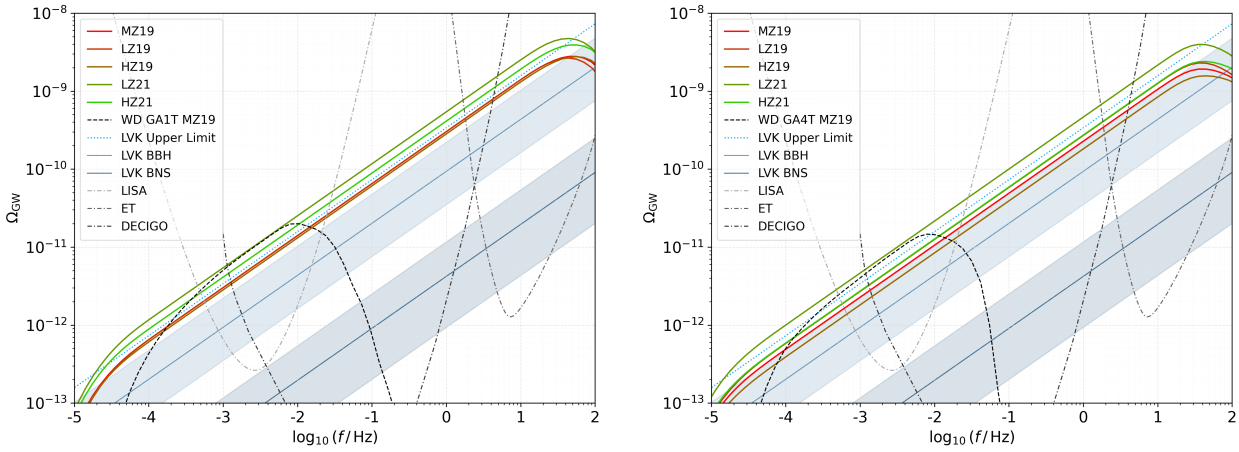
Figure 7 shows the GWB curves of all four mass transfer models under SFRD MZ19, chosen to isolate the mass transfer impact. The curve follows the same structure as the WD curve, which consists of three parts. It starts with the expected $f^{10/3}$ dependence in the static regime, where binaries evolve too slow and never make it to their maximum frequency. Secondly the curve transitions to the $f^{2/3}$ inspiral dependence, which was predicted when assuming a

constant stream of systems going into every next frequency bin. Finally the curves shows an exponential drop-off due to the ISCO cutoff at which we arbitrarily chose to stop calculating, and the maximum $\nu_{\rm min}$ of the population. The results sit slightly above the LVK uncertainty band for BHB and close to the LVK upper limit. The main takeaway is that the difference between the α and γ formalisms for the first mass transfer phase is negligible, and that $\alpha = 4$ yields a slightly lower amplitude than $\alpha = 1$.



(a) AA1

(b) AA4



(c) GA1

(d) GA4

Figure 8: GWB curves across all five SFRD models for each mass transfer prescription. An example using SFRD MZ19 shows how these lines are built up from the 6 metallicities in Appendix B.

In Figure 8 we find all combinations of mass transfer with the SFRD model. The trend established in Section 2.4 is clearly visible, showing that lower metallicity yields a higher merger rate and thus a higher amplitude. LZ21 consistently produces the highest amplitude across all mass transfer models, while HZ21 follows as the next highest, consistent with the higher

metallicity assumed in the 2021 models. The three 2019 models are nearly indistinguishable from one another. Comparing Figures 7 and 8, the $\alpha = 1$ versus $\alpha = 4$ distinction has the largest impact on the amplitude, with LZ21 being the only SFRD model that produces a notably higher curve.

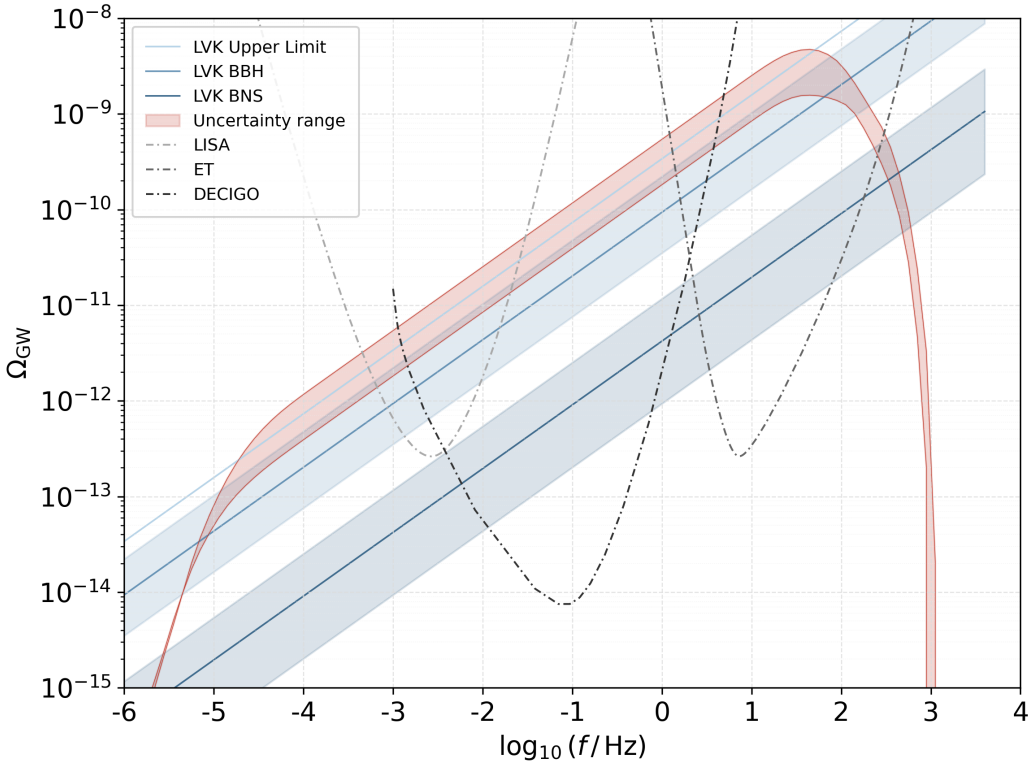


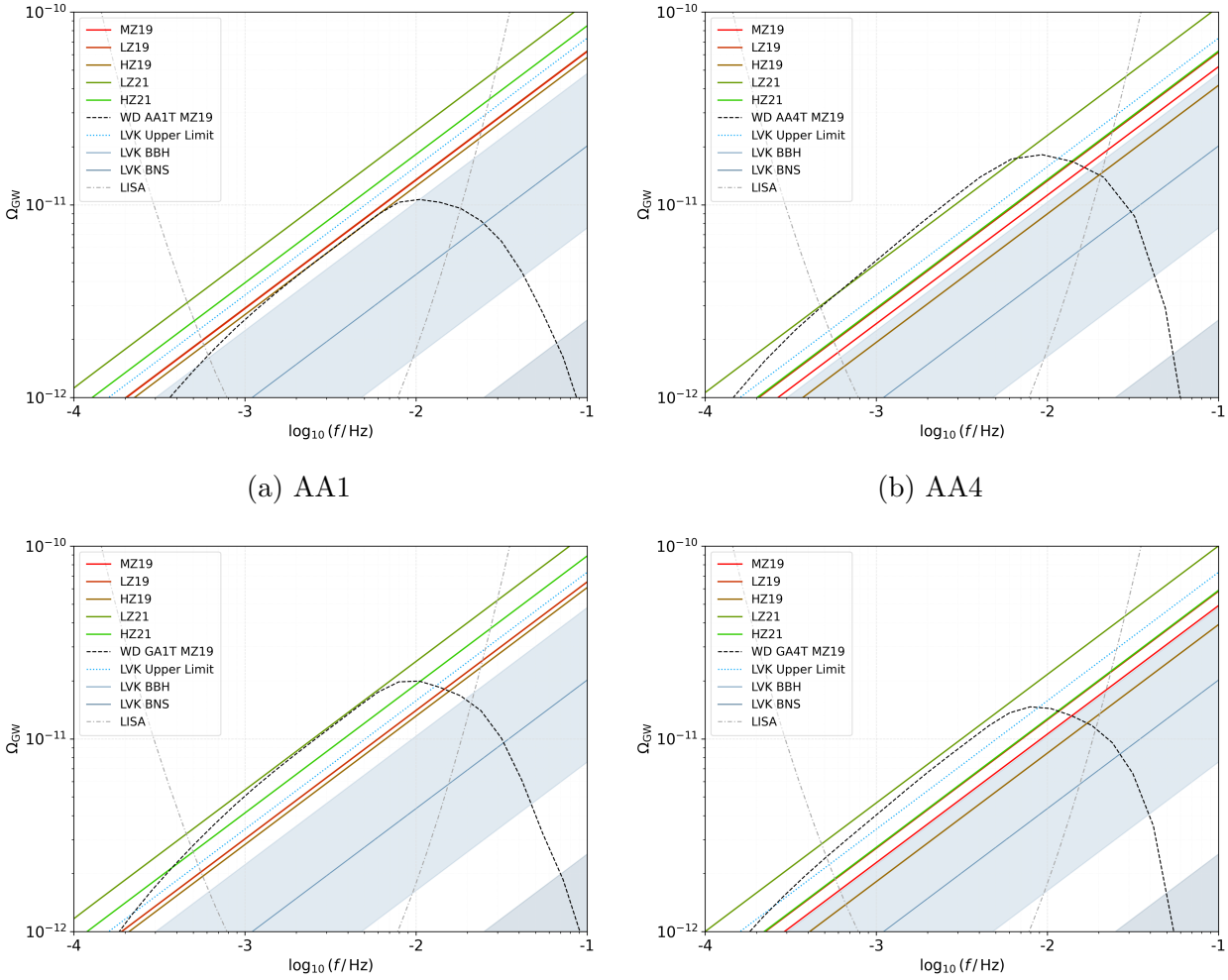
Figure 9: Total uncertainty range of the BHB GWB in this thesis, bounded by the highest and lowest amplitude SFRD and mass transfer model combinations.

Figure 9 shows the combined uncertainty range from all model combinations. The centre of the uncertainty band falls close to the LVK upper limit, while the lower bound just reaches into the LVK BBH uncertainty range, indicating that our results predict a relatively high but not unreasonable BHB merger rate. Now that we have verified the overall characteristics of the different curves, it is worth looking more into the differences within the LISA sensitivity range. To verify the $f^{2/3}$ dependence within the LISA band, a power law fit to the middle values of the curve in Figure 9 gives

$$\Omega_{\mathrm{GW}}(f) = 3.620 \times 10^{-10} \cdot f^{0.666716}, \tag{3.1}$$

confirming the $f^{2/3}$ inspiral dependence to high accuracy, and showing that BHB systems enter the inspiral regime much more quickly than WD binaries, as can also be seen in Figure 10 and [Hofman and Nelemans, 2024].Within the zoomed-in view of the LISA band it also becomes more clearly visible how the BHB curve competes with the WD curve. While we had concluded

that the γ and α formalisms play no significant role in the amplitude of the BHB background, they do affect the WD background, and this appears to fundamentally change which of the two dominates within the LISA band, as seen in Figures 10a and 10c. For the $\alpha = 4$ formalism we found that the amplitude of the BHB background is lowered, increasing the probability of WD domination, as shown in Figures 10b and 10d. It must be noted that only in the AA1 case do all BHB curves compete with the WD background, and keeping in mind that in [Hofman and Nelemans, 2024] the WD AA1 MZ19 result represents only one of the middle models, the actual curve may still be higher, further suppressing the relative contribution of the lower BHB models. In all other three cases discussed in this thesis, only LZ21 and in one instance HZ21 approach the WD results, while the remaining models fall significantly below.



(a) AA1

(b) AA4

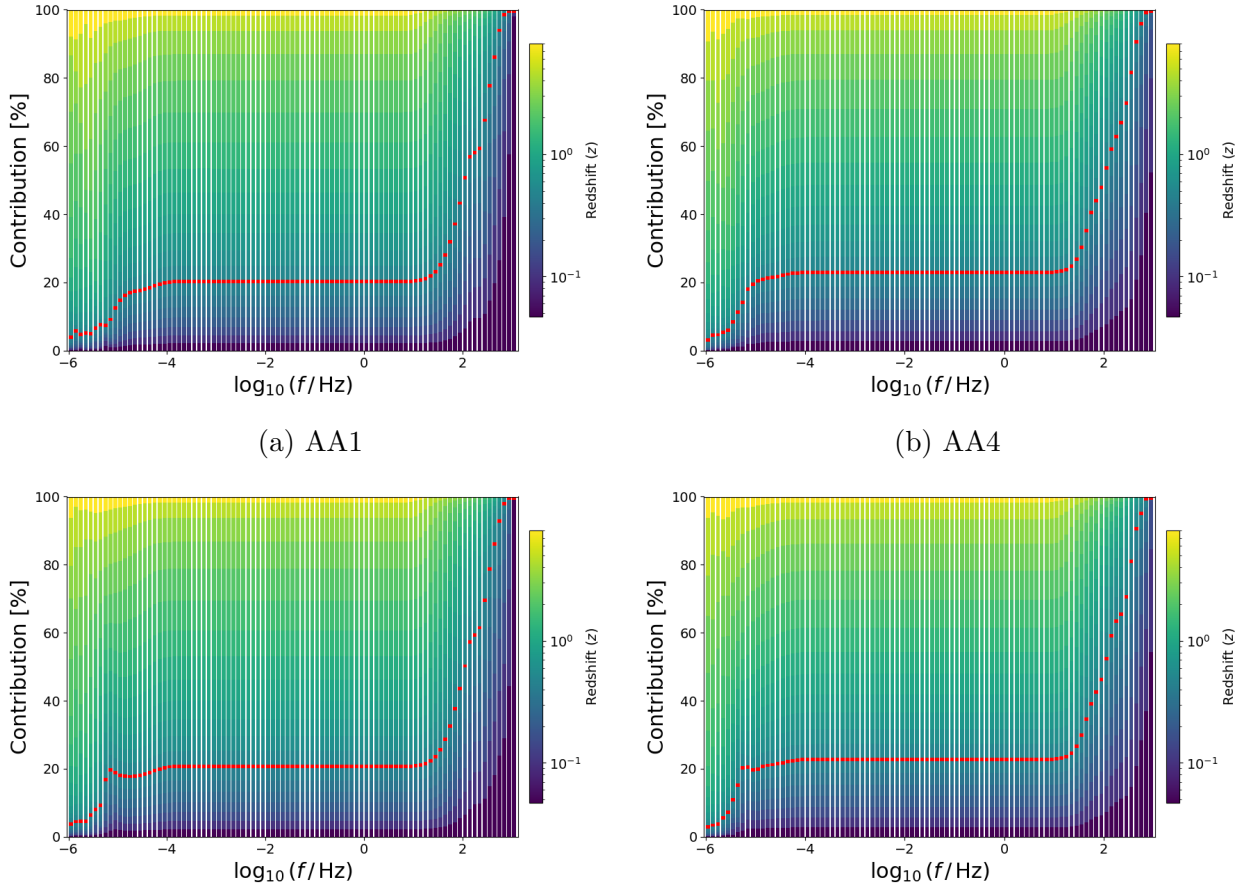
(c) GA1

(d) GA4

Figure 10: GWB curves across all five SFRD models for each mass transfer prescription, zoomed in on the LISA sensitivity range.

3.2 Model Distributions

To further investigate the origins of the main results we turn to the specifics of the different populations, starting with the redshift contributions.



(a) AA1

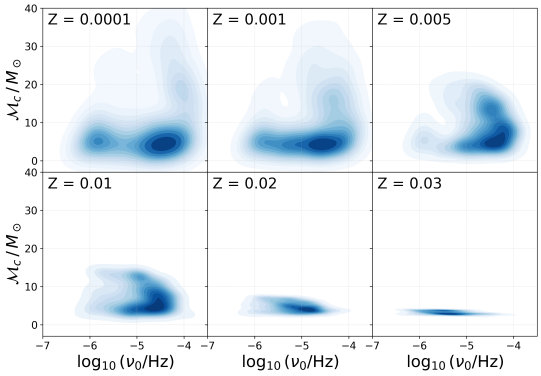
(b) AA4

(c) GA1

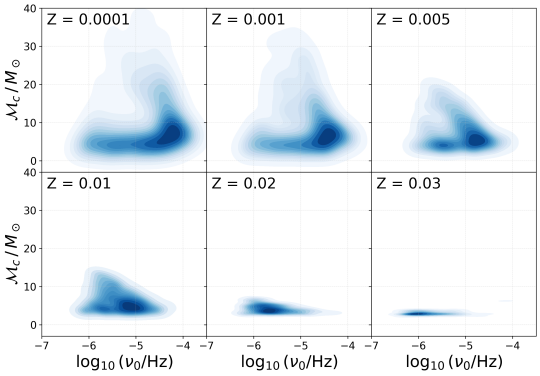
(d) GA4

Figure 11: Contributions of all redshift bins to the total GWB as a function of frequency, for all mass transfer models at SFRD MZ19. The red threshold indicates z = 0.43.

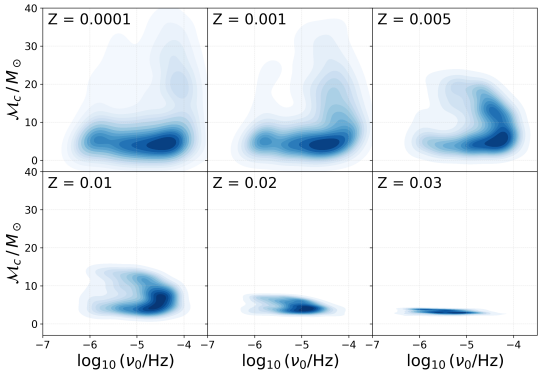
Figure 11 confirms that integrating up to $z = 8$ is sufficient, as the highest redshift bins contribute negligibly to the total signal. Nearly half of the contribution in the inspiral phase comes from within $z = 1$, while the bins nearest to merger are dominated by the lowest redshifts. No significant difference between the mass transfer models is visible. We next consider the initial population variables from the population synthesis, namely the chirp masses, formation times, initial orbital frequencies, and the maximum frequency.



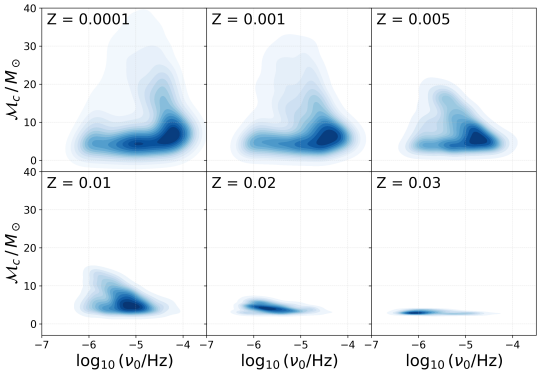
(a) AA1



(b) AA4



(c) GA1

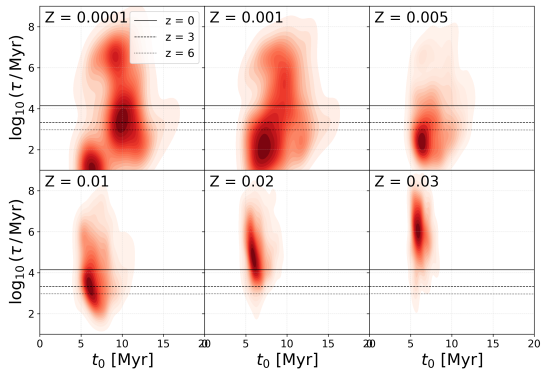


(d) GA4

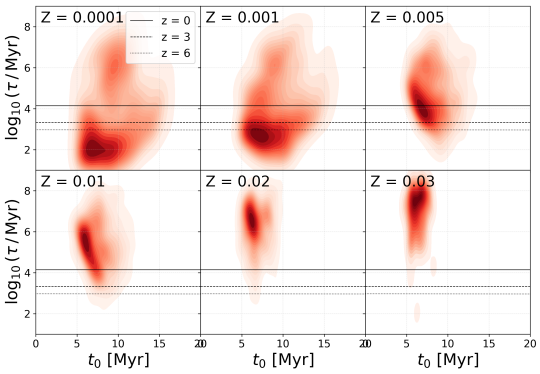
Figure 12: Chirp mass plotted against initial orbital frequency ν_0 for each mass transfer model, the density color-coded.

The density plots in Figure 12 show how the initial populations differ in M_c versus ν_0 across the different model combinations. The impact of metallicity is clearly visible: higher metallicity populations lack the most massive systems, a direct consequence of stronger stellar winds preventing stars from retaining enough mass to produce high chirp mass binaries. Since $\Omega_{\mathrm{GW}} \propto M_c^{5/3}$, this directly suppresses the contribution of high metallicity populations to the background. The ν_0 distributions show no significant differences between models.

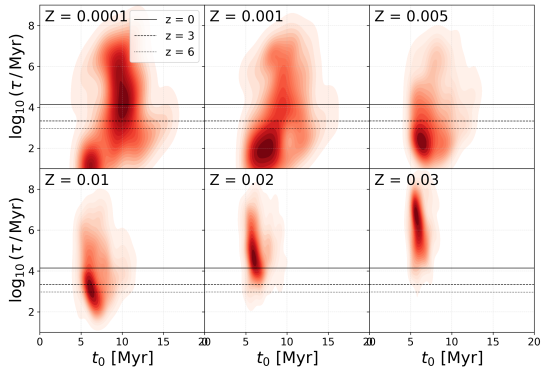
Comparing the $\gamma\alpha$ and $\alpha\alpha$ populations, the two appear nearly identical, consistent with the earlier finding that the choice of first mass transfer formalism has negligible impact on the resulting background. Some difference between $\alpha = 1$ and $\alpha = 4$ is however visible. For the $\alpha = 4$ case there appears to be a slightly larger subgroup of higher chirp mass systems at $z = 0.0001$ and $z = 0.001$. This is somewhat counterintuitive given that $\alpha = 4$ produces a slightly lower AGWB amplitude, suggesting this excess does not translate into a higher background, possibly due to differences in merger timescales and the total impact of the other metallicity groups.



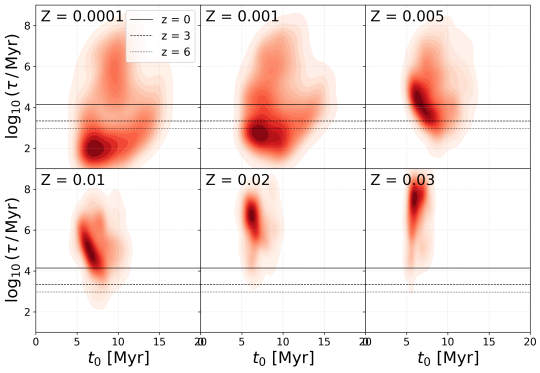
(a) AA1



(b) AA4



(c) GA1

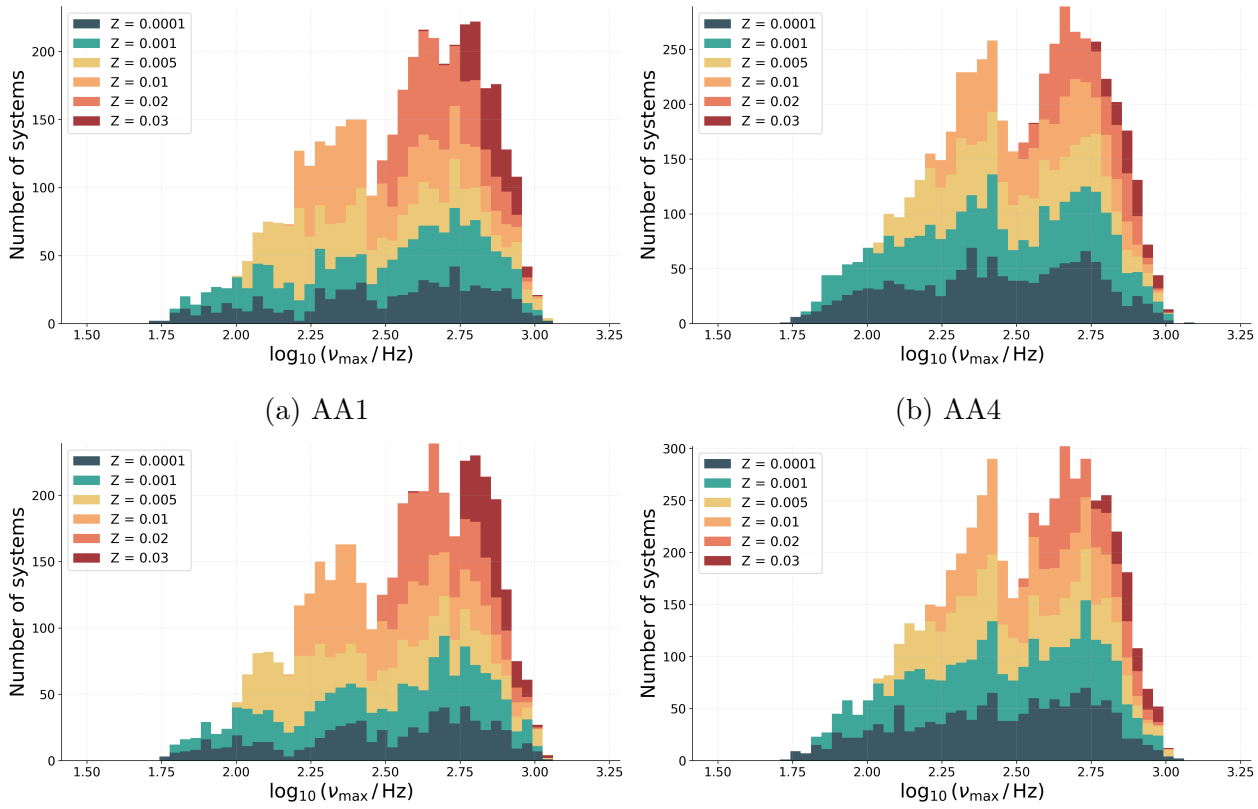


(d) GA4

Figure 13: Merger timescale τ plotted against formation time t_0 for each mass transfer model, the density color-coded. The three horizontal lines indicate the Hubble time at three different redshifts, separating the static and inspiral regimes. These lines are horizontal because only the τ axis is logarithmic, making t_0 effectively irrelevant to this boundary.

Figure 13 shows the merger timescale τ, computed via Eq. 1.16, plotted against formation time t_0 for each mass transfer model. The horizontal lines indicate the Hubble time boundary, above which systems fall into the static regime and never contribute to the inspiral part of the background. Higher metallicity populations are clearly more concentrated in this regime, consistent with their lower chirp masses and wider orbits from stronger stellar winds, both of which increase τ.

The difference between $\alpha = 1$ and $\alpha = 4$ is most clearly visible here: the wider post-CEE orbits produced by $\alpha = 4$ push more systems into the static regime, directly explaining the lower $\Omega_{\rm GW}$ amplitude seen in Figures 7 and 8, with higher metallicity populations being most strongly affected. The $\gamma\alpha$ and $\alpha\alpha$ populations again show no significant difference.



(a) AA1

(b) AA4

(c) GA1

(d) GA4

Figure 14: Stacked bar charts of the number of systems per $\nu_{\max}$ bin for each mass transfer model, colour-coded by metallicity.

Figure 14 shows the $\nu_{\max}$ distribution of the populations, which sets the upper frequency cutoff for each binary's contribution to the background. The $\alpha = 4$ case shows relatively more low metallicity systems, the $\alpha = 1$ case on the other hand shows more high metallicity systems surviving to merger, as expected from their narrower post-CEE orbits which place fewer systems above the Hubble time boundary. Combining this finding with the previous section, onde can deduce that the higher AGWB for $\alpha = 1$ mainly comes from less static and a higher volume higher metallicity systems.

Across all models, the metallicity bins form an approximate bell curve in $\nu_{\max}$, with higher metallicity systems peaking at higher $\nu_{\max}$. This is a direct consequence of their lower chirp masses, which correspond to less massive and more compact binaries that can reach closer separations before ISCO contact is reached. Conversely, the most massive low metallicity systems reach ISCO at relatively wider separations, shifting their $\nu_{\max}$ distribution towards lower frequencies. The overall spread in $\nu_{\max}$ across metallicity bins therefore reflects the same mass dependence that drives the metallicity trends seen throughout the previous figures.

4 Discussion and Conclusion

In this thesis we have built on the framework of [Staelens and Nelemans, 2024] and [Hofman and Nelemans, 2024] to construct a GWB for BHB systems. We first verified the circular orbit assumption and demonstrated that the effect of eccentricity on Ω_{GW} is negligible, consistent with the findings of [Perigois et al., 2021], which used StarTrack as their population synthesis code rather than SeBa. Regarding the choice of SFRD, we showed that the model of [Madau and Dickinson, 2014] produces unphysical results for this application, and that a metallicity-dependent SFRD is required, building on the approach of [Hofman and Nelemans, 2024]. For black hole binaries specifically, metallicity is the most important parameter, as higher metallicity drives stronger stellar winds, resulting in lower chirp masses and more systems in the static regime, both of which suppress the amplitude of Ω_{GW}.

The resulting GWB curves sit on the high end of the [LIGO Scientific Collaboration et al., 2025] estimated range, and comparison with similar studies confirms this trend: our results are a factor of two to three higher than both [Li et al., 2023], which uses COSMIC as their population synthesis code, and the aforementioned [Perigois et al., 2021]. This relatively high merger rate should however be interpreted in the context of the limited parameter space explored here. Only two parameters have been varied in this work, whereas common envelope efficiency, supernova kicks, and mass accretion prescriptions also have a notable impact on the final population, as demonstrated by [Santoliquido et al., 2020] across a wide range of model variations, and by [Dorozsmai and Toonen, 2024] in more detail specifically for SeBa. Most importantly, both studies find that the α formalism has the largest single impact on the merger rate, with the full range of α values spanning orders of magnitude in the resulting merger rate. This is consistent with our results, in which $\alpha = 4$ yields a lower amplitude than $\alpha = 1$.

Significant differences can also arise from the choice of population synthesis code itself, independently of the assumed parameters. [Gallegos-Garcia et al., 2021] shows that merger rates between COSMIC and MESA can differ by up to two orders of magnitude, while [Dominik et al., 2012] finds that SeBa produces more massive BHB systems with a lower merger rate than StarTrack, though the two effects may partially cancel in the total Ω_{GW}.

Within the LISA sensitivity band, the BHB AGWB follows the $f^{2/3}$ dependence throughout, while some of the higher amplitude cases are able to compete with the WD curve from [Hofman and Nelemans, 2024]. In the majority of results however the WD background appears dominant in the LISA range, at least when comparing to the medium metallicity and medium amplitude cases, further supporting the findings of [Hofman and Nelemans, 2024] and [Staelens and Nelemans, 2024] which conclude that WD binaries are the dominant source of gravitational waves in the LISA frequency band.

In conclusion, the true uncertainty on the BHB GWB is considerably larger than shown in Figure 9, and the two parameters tested here represent only a subset of the relevant physics. The α value for the CEE phase has a clear and measurable impact on the merger rate, while the distinction between the $\gamma\alpha$ and $\alpha\alpha$ mass transfer prescriptions appears negligible for BHB

systems. Since SeBa was originally developed for less massive systems such as WD binaries, it is possible that certain untested parameters allow many high mass systems to survive that would not have done so in another population synthesis code, making the results in this work significantly higher than the LVK estimates would suggest. Black hole binary evolution remains an active area of research, and the full uncertainty on the GWB from BHB reflects this. Further work comparing the most impactful parameters like metallicity, CEE efficiency, and the secondary parameters identified in [Dorozsmai and Toonen, 2024] and [Santoliquido et al., 2020] across multiple population synthesis codes, in combination with improved black hole evolution models and future LISA observations, will be essential to better constrain the BHB merger rate and its contribution to the AGWB.

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A General Relativity

A.1 Gravitational Wave Equation

We want to find a solution for the field equations in vacuum, meaning that

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = 8\pi G\, T_{\mu\nu} = 0. \tag{A.1}$$

We take a linearized Ansatz as an approach using $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, expanded to first order. The Christoffel symbols are

$$\Gamma^{\lambda}_{\mu\nu} = \frac{1}{2} g^{\lambda\alpha} \left(\partial_\mu g_{\alpha\nu} + \partial_\nu g_{\alpha\mu} - \partial_\alpha g_{\mu\nu} \right). \tag{A.2}$$

Plugging in our $g_{\mu\nu}$ and discarding terms higher than first order gives the linearised Christoffel symbols

$$\delta\Gamma^{\lambda}_{\mu\nu} = \frac{1}{2} \eta^{\lambda\alpha} \left(\partial_\mu h_{\alpha\nu} + \partial_\nu h_{\alpha\mu} - \partial_\alpha h_{\mu\nu} \right). \tag{A.3}$$

The Ricci tensor and scalar are defined by

$$R_{\mu\nu} = R^{\alpha}{}_{\mu\alpha\nu}, \qquad R = g^{\mu\nu} R_{\mu\nu}, \tag{A.4}$$

for which the Riemann tensor is needed,

$$R^{\beta}{}_{\mu\alpha\nu} = \partial_\alpha \Gamma^{\beta}_{\mu\nu} - \partial_\nu \Gamma^{\beta}_{\mu\alpha} + \Gamma^{\beta}_{\gamma\alpha} \Gamma^{\gamma}_{\mu\nu} - \Gamma^{\beta}_{\gamma\nu} \Gamma^{\gamma}_{\mu\alpha}. \tag{A.5}$$

For the linearised Ricci tensor we retain only the linear terms in Eq. A.5,

$$\delta R_{\mu\nu} = \partial_\alpha \delta\Gamma^{\alpha}_{\mu\nu} - \partial_\nu \delta\Gamma^{\alpha}_{\mu\alpha}, \tag{A.6}$$

and expanding using Eq. A.3 gives

$$\begin{aligned} \delta R_{\mu\nu} = {} & \frac{1}{2} \left(\partial_\alpha \partial_\mu h^{\alpha}{}_{\nu} + \partial_\alpha \partial_\nu h^{\alpha}{}_{\mu} - \partial_\alpha \partial^\alpha h_{\mu\nu} - \partial_\mu \partial_\nu h^{\alpha}{}_{\alpha} \right) \\ & - \frac{1}{2} \left(\partial_\nu \partial_\mu h^{\alpha}{}_{\alpha} + \partial_\nu \partial_\alpha h^{\alpha}{}_{\mu} - \partial_\nu \partial_\alpha h^{\alpha}{}_{\mu} \right). \end{aligned} \tag{A.7}$$

The linearised Ricci scalar is then obtained by contracting with $\eta^{\mu\nu}$,

$$\begin{aligned} \delta R &= \eta^{\mu\nu} \delta R_{\mu\nu} \\ &= \frac{1}{2} \left(\partial_\alpha \partial^\mu h^{\alpha}{}_{\mu} + \partial_\alpha \partial^\nu h^{\alpha}{}_{\nu} - \partial_\alpha \partial^\alpha h^{\mu}{}_{\mu} - \partial^\mu \partial_\mu h^{\alpha}{}_{\alpha} \right) \\ &\quad - \frac{1}{2} \left(\partial^\nu \partial_\nu h^{\alpha}{}_{\alpha} + \partial^\nu \partial_\alpha h^{\alpha}{}_{\nu} - \partial^\nu \partial_\alpha h^{\alpha}{}_{\nu} \right) && \text{(A.8)} \\ &= \frac{1}{2} \left(\partial_\alpha \partial^\mu h^{\alpha}{}_{\mu} + \partial_\alpha \partial^\nu h^{\alpha}{}_{\nu} - \Box h^{\alpha}{}_{\alpha} - \Box h^{\alpha}{}_{\alpha} \right) \\ &\quad - \frac{1}{2} \left(\Box h^{\alpha}{}_{\alpha} + \partial^\nu \partial_\alpha h^{\alpha}{}_{\nu} - 0 \right) && \text{(A.9)} \\ &= \partial_\alpha \partial^\mu h^{\alpha}{}_{\mu} - \Box h, && \text{(A.10)} \end{aligned}$$

where $\Box = \partial^\alpha \partial_\alpha$ is the d'Alembertian and $h \equiv h^\alpha{}_\alpha$ is the trace of the metric perturbation. Combining the linearised Ricci tensor A.7 and scalar A.8 into the linearised Einstein tensor and setting it to zero,

$$\begin{aligned}\delta G_{\mu\nu} &= \delta R_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}\,\delta R = 0 \\ &= \frac{1}{2}\left(\partial_\alpha \partial_\mu h^\alpha{}_\nu + \partial_\alpha \partial_\nu h^\alpha{}_\mu - \Box h_{\mu\nu} - \partial_\mu \partial_\nu h^\alpha{}_\alpha\right) \\ &\quad - \frac{1}{2}\eta_{\mu\nu}\left(\partial_\alpha \partial^\mu h^\alpha{}_\mu - \Box h\right) = 0.\end{aligned}$$

To find solutions we make a perturbation matrix

$$h_{\mu\nu} = \begin{pmatrix} -2\Phi & w_1 & w_2 & w_3 \\ w_1 & 2s_{11} - 2\Psi & 2s_{12} & 2s_{13} \\ w_2 & 2s_{12} & 2s_{22} - 2\Psi & 2s_{23} \\ w_3 & 2s_{13} & 2s_{23} & 2s_{33} - 2\Psi \end{pmatrix}, \tag{A.11}$$

of which the reasoning of this composition will become more clear in the next section, whereas the factors 2 that appear are conventional to retrieve newtonian gravity and to fit nicely into the Einstein equation.

A.2 SVT Decomposition

The metric perturbation $h_{\mu\nu}$ has by definition 10 degrees of freedom (DOF), arising from the symmetry of a 4×4 matrix. With 4 gauge freedoms corresponding to the 4 spacetime dimensions, we are left with 6 physical degrees of freedom. These are captured by the scalar-vector-tensor (SVT) decomposition, where each sector describes different physics.

The scalar sector, described by Φ and Ψ, accounts for 2 DOF and recovers Newtonian gravity in the non-relativistic limit, where Φ plays the role of the gravitational potential and Ψ describes the spatial curvature induced by matter density. The vector sector, described by w_i, starts with 3 DOF and is associated with gravitomagnetism, the gravitational analogue of magnetism arising from mass currents. Since the longitudinal part of w_i can always be written as the gradient of a scalar, it carries no independent physical information and we may impose the transverse condition

$$\partial^i w_i = 0, \tag{A.12}$$

which filters out the frame dragging contribution and uses 1 DOF, leaving 2 independent transverse directions. The tensor sector, described by s_{ij}, accounts for the remaining 5 DOF. Gravitational waves distort spacetime rather than inflate it, so s_{ij} must be traceless,

$$s^i{}_i = 0, \tag{A.13}$$

and transverse,

$$\partial^i s_{ij} = 0, \tag{A.14}$$

by the same argument as for the vector sector. This uses 3 more DOF, leaving exactly 2 DOF to describe the two polarization states of gravitational waves, giving a total of $2+2+2=6$ physical DOF.

We need s_{ij} to be traceless as imposed in Eq. A.13. Computing the trace of the spatial part of the perturbation matrix Eq. A.11,

$$\mathrm{Tr}(h_{ij}) = \mathrm{Tr}(2s_{ij} - 2\Psi\delta_{ij}) = 0 - 2\Psi\delta^{i}{}_{i} = -6\Psi. \tag{A.15}$$

By taking this trace explicitly, we can identify the scalar spatial curvature Ψ as

$$\Psi = -\frac{1}{6}\delta^{ij}h_{ij}. \tag{A.16}$$

This identification shows that Ψ represents the isotropic (trace) degree of freedom in the spatial metric. We then substitute the full decomposition into the linearised vacuum Einstein equations, $\delta G_{\mu\nu} = 0$. Under the imposition of the transverse gauge conditions Eq. A.12 and Eq. A.14, the components of the field equations decouple into the following constraints:

$$\delta G_{00} = 2\nabla^2\Psi = 0 \implies \Psi = 0, \tag{A.17}$$

$$\delta G_{0i} = \frac{1}{2}\nabla^2 w_i = 0 \implies w_i = 0, \tag{A.18}$$

$$\delta G^{i}{}_{i} = \nabla^2(\Phi - \Psi) = 0 \implies \Phi = \Psi = 0. \tag{A.19}$$

Because the vacuum equations $\nabla^2\Psi = 0$ and $\nabla^2 w_i = 0$ have only the trivial solution in asymptotically flat spacetime, the scalar potentials Φ, Ψ and the vector w_i must vanish. This leaves only the space-space part of Eq. **??**,

$$\delta G_{ij} = -\frac{1}{2}\Box s_{ij} = 0 \implies \Box s_{ij} = 0. \tag{A.20}$$

In the transverse-traceless (TT) gauge the perturbation matrix Eq. A.11 then reduces to

$$h^{TT}_{\mu\nu} = \begin{pmatrix} 0 & 0 \\ 0 & 2s_{ij} \end{pmatrix}, \tag{A.21}$$

where $\Phi = \Psi = 0$ and $w_i = 0$ set the time-time and time-space components to zero. The transverse condition Eq. A.14 then sets the z-row and z-column to zero for a wave propagating in the z-direction, and subsequently the traceless condition Eq. A.13 requires $s_{22} = -s_{11}$. We then make the plane wave Ansatz to satisfy Eq. A.20,

$$h^{TT}_{\mu\nu} = C_{\mu\nu}\, e^{ik_\alpha x^\alpha}, \tag{A.22}$$

where $C_{\mu\nu}$ is the polarization tensor, symmetric by definition, set to be transverse and traceless,

$$C_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & h_+ & h_\times & 0 \\ 0 & h_\times & -h_+ & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \tag{A.23}$$

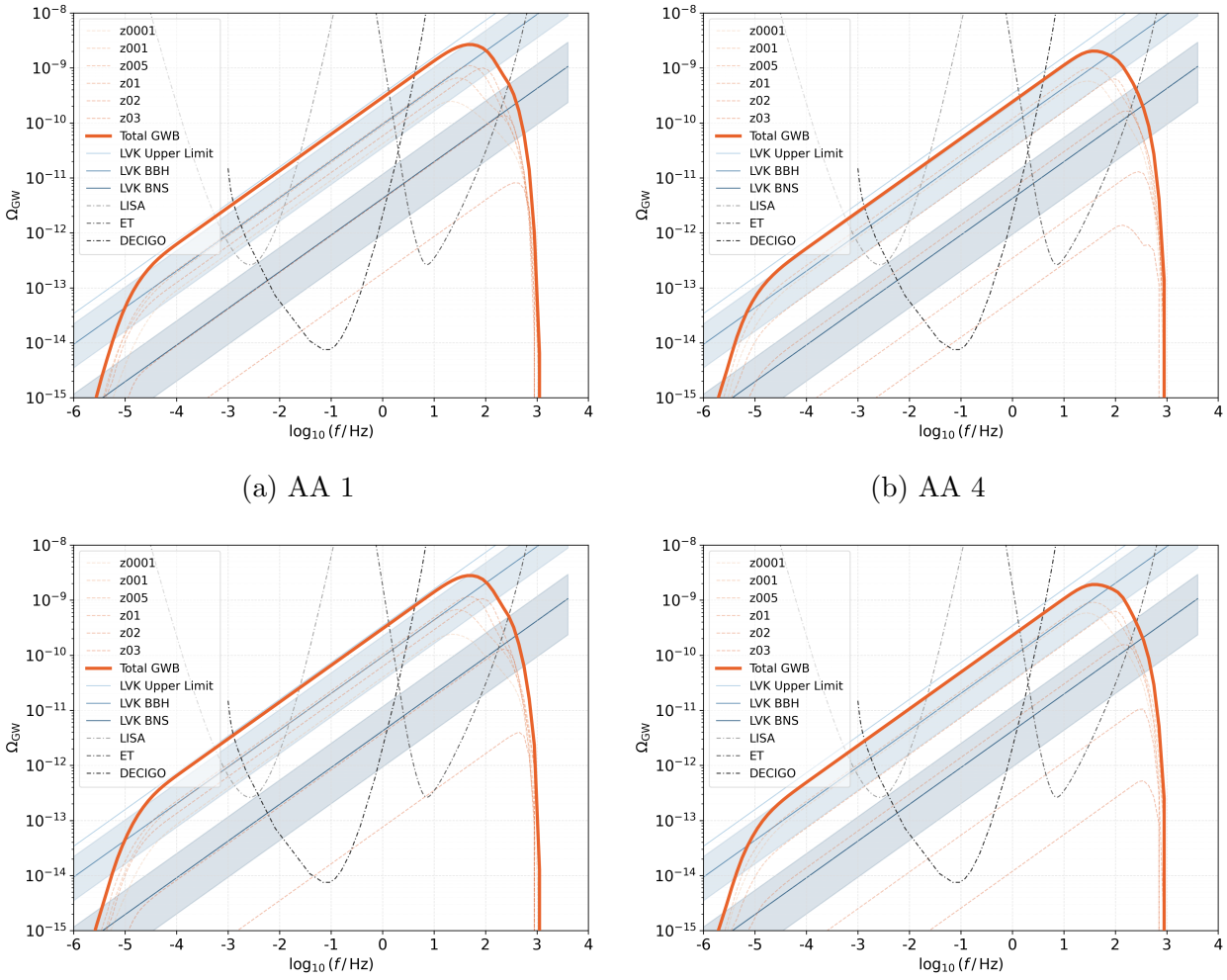
where h_+ and $h_\times$ represent the 2 remaining DOF as the two polarization states of gravitational waves. Substituting the Ansatz Eq. A.22 into $\Box\, h^{TT}_{\mu\nu} = 0$ gives the dispersion relation

$$\omega^2 - k_i^2 = 0, \tag{A.24}$$

meaning gravitational waves propagate at the speed of light. The transverse condition Eq. A.14 further requires

$$k^\mu C_{\mu\nu} = 0. \tag{A.25}$$

B Additional Figures



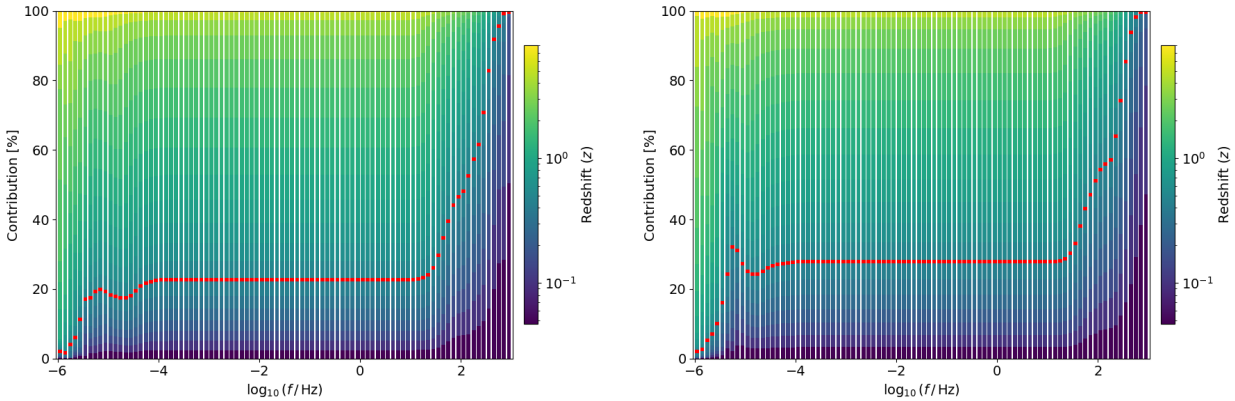
(a) AA 1

(b) AA 4

(c) GA 1

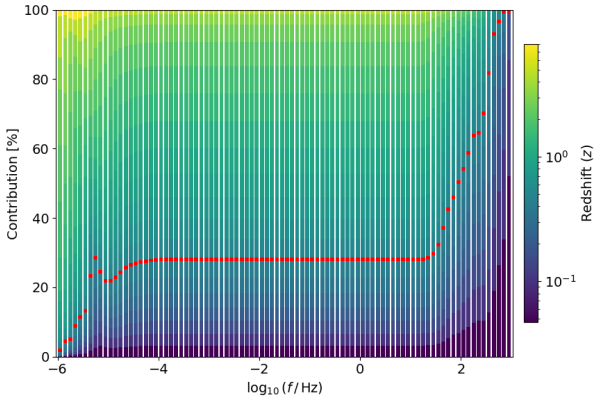
(d) GA 4

Figure 15: Buildup of the AGWB by 6 metallicities using MZ19, integrated over time.

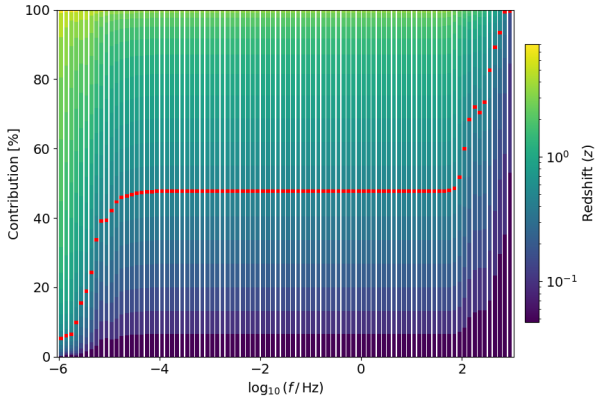


(a) z0001

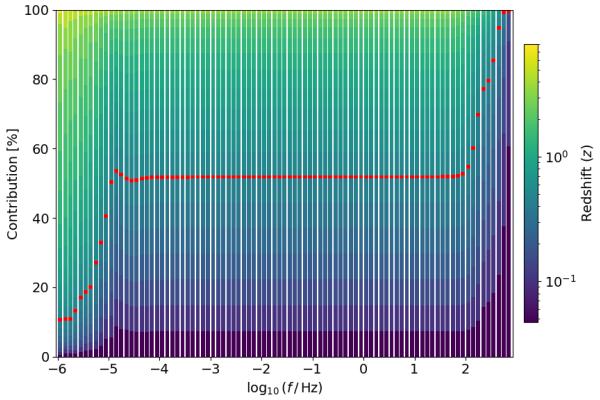
(b) z0005



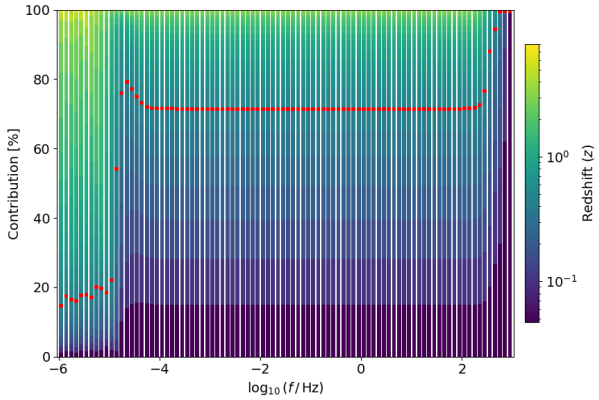
(c) z001



(d) z005

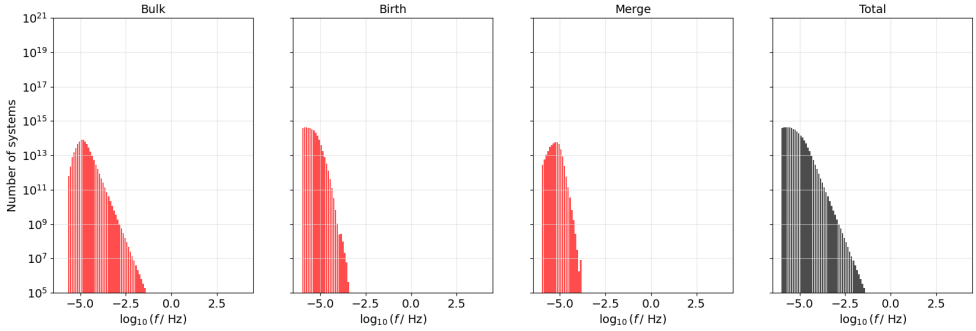


(e) z01

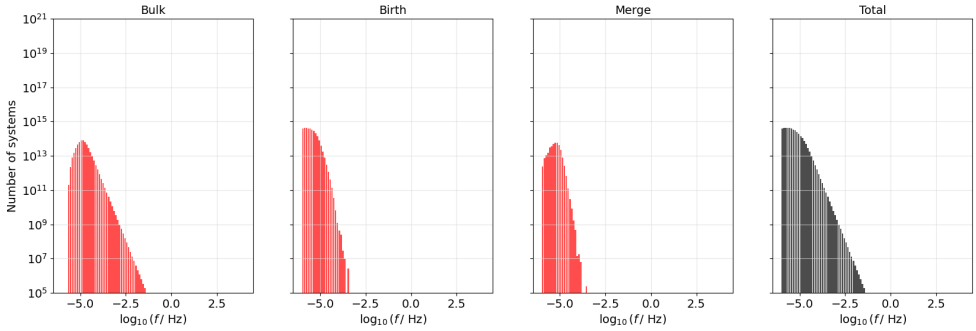


(f) z02

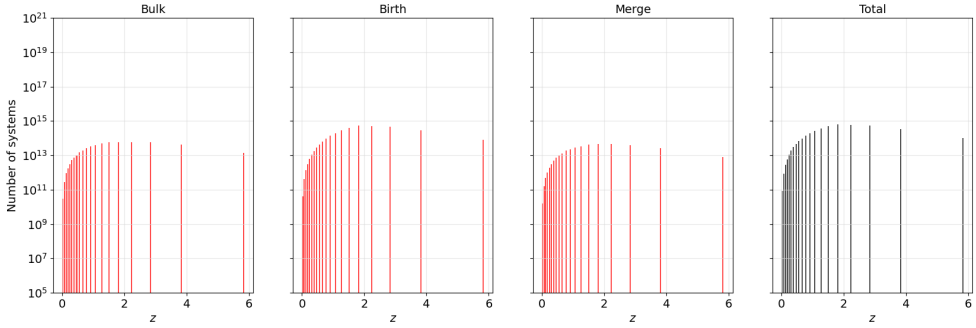
Figure 16: The z contribution results for a GA4 population integrated over time, when using [Madau and Dickinson, 2014] as SFH for 6 different individual runs. A clear trend can be seen resulting in more contributions from closer binaries at higher metallicities.



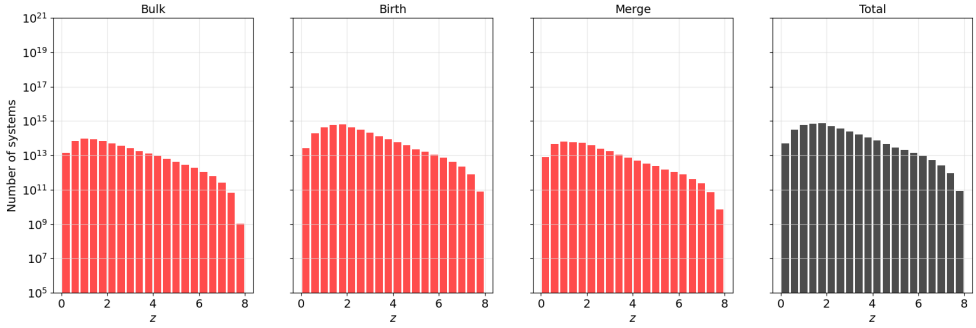
(a) Frequency distribution for time integration.



(b) Frequency distribution for redshift integration.

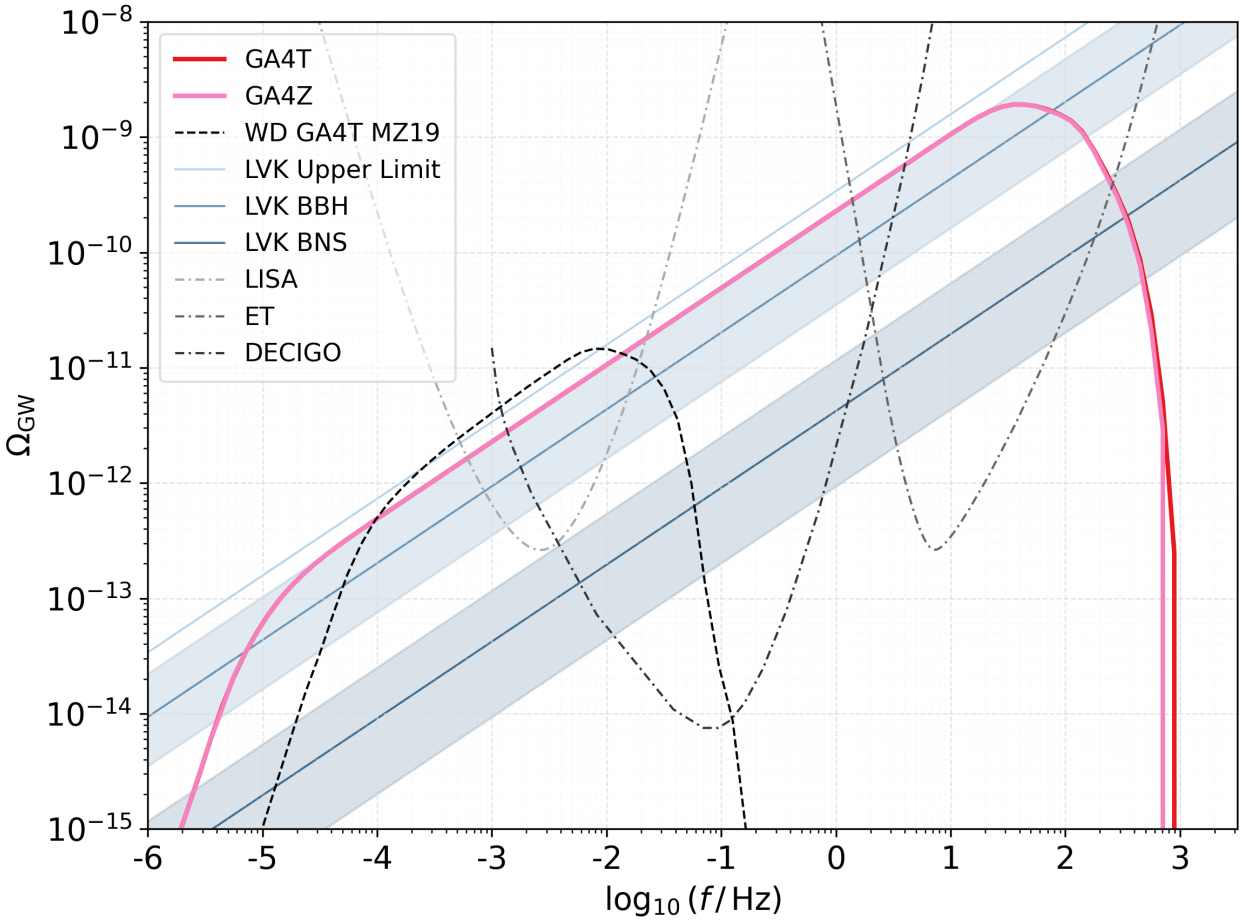


(c) Redshift distribution for time integration.

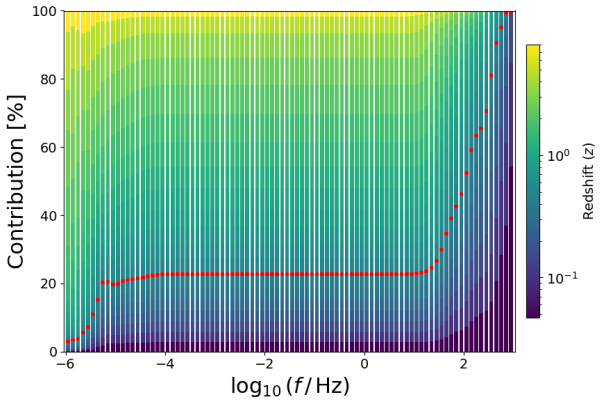


(d) Redshift distribution for redshift integration.

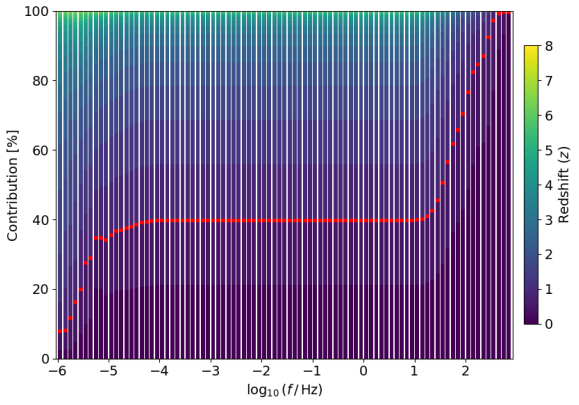
Figure 17: Different ways the models are distributed when integrating over time and redshift.



(a) Resulting AGWB curve for a GA4 MZ19 model when integrated via redshift and time, showing negligible difference between the two.



(b) Z contribution when integrated over time.



(c) Z contribution when integrated over redshift.

Figure 18: Results of 2 ways of integrating our models, showing a slight difference in z contribution for the 2, but resulting in no difference for the AGWB. The red threshold in the z contribution figures shows z=0.43, while the two figures have different contribution bins.

C Data and Code

The code and used data for this work can be found at `https://github.com/akoshkardos/grav_wav_background`.
The original code developed for white dwarf binaries was taken from `https://github.com/SeppeStaelens/White_Dwarf_AGWB`.